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MASTER'S THESIS

From Internal Representations to Output Distributions: Improving Reliability, Control, and Training Efficiency in Large Language Models

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МАГИСТЕРСКАЯ ДИССЕРТАЦИЯ

От внутренних представлений до выходных распределений: повышение надёжности, управляемости и эффективности обучения больших языковых моделей

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From Internal Representations to Output Distributions: Improving Reliability, Control, and Training Efficiency in Large Language Models

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ABSTRACT

Large language models (LLMs) have demonstrated remarkable capabilities across diverse tasks, yet their deployment faces three critical challenges: unreliable outputs that may contain errors or hallucinations, limited control over generation behavior, and inefficient training procedures that waste computational resources on redundant data. This thesis addresses these challenges through a unified approach that leverages signals from the model's forward pass—both internal representations and output distributions—to improve reliability, control, and training efficiency.

We present three interconnected studies. First, we investigate uncertainty estimation by analyzing the relationship between token-level entropy and answer correctness. Our experiments on MMLU-Pro across four models (Phi-4, Mistral-Small-24B, Qwen-1.5B, Qwen-72B) demonstrate that single-token answer entropy achieves ROC-AUC scores up to 0.83 for predicting incorrect answers in knowledge-dependent domains, while Model-as-Judge approaches perform near-randomly. Second, we explore the geometric structure of multilingual representations in LLM latent space, discovering that language-specific directions can be identified through PCA with 96–100% classification accuracy across four language pairs. By projecting out the source language component from hidden states, we reduce distributional divergence by up to 55% and achieve 63–95% reduction in Code-Switching Index in generated text, without requiring additional training. Third, we develop a complexity-aware fine-tuning pipeline that uses entropy to stratify training data and applies targeted interventions: direct supervised fine-tuning for regular examples and chain-of-thought distillation for complex ones. This approach achieves accuracy improvements from 0.39 to 0.52 (Qwen-3B) and 0.51 to 0.64 (Phi-4-Mini) on MMLU-Pro while using 79–82% less training data, with results generalizing to GSM8K and MedMCQA datasets.

Together, these contributions demonstrate that accessible signals from a single forward pass provide valuable information for addressing fundamental challenges in LLM deployment. The methods developed require no architectural modifications or expensive retraining, making them practical for real-world applications.

Keywords: large language models, uncertainty estimation, latent space steering, complexity-aware fine-tuning, internal representations

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От внутренних представлений до выходных распределений: повышение надёжности, управляемости и эффективности обучения больших языковых моделей

Андрей Гончаров

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Аннотация

Большие языковые модели (LLM) продемонстрировали выдающиеся способности в решении разнообразных задач, однако их внедрение сталкивается с тремя критическими проблемами: ненадёжные выходные данные, которые могут содержать ошибки или галлюцинации, ограниченный контроль над генерацией и неэффективные процедуры обучения, расходующие вычислительные ресурсы на избыточные данные. Данная диссертация решает эти проблемы с помощью единого подхода, использующего сигналы прямого прохода модели — как внутренние представления, так и выходные распределения — для повышения надёжности, управляемости и эффективности обучения.

Мы представляем три взаимосвязанных исследования. Во-первых, мы исследуем оценку неопределённости путём анализа связи между энтропией на уровне токенов и правильностью ответов. Наши эксперименты на MMLU-Pro с четырьмя моделями (Phi-4, Mistral-Small-24B, Qwen-1.5B, Qwen-72B) показывают, что энтропия ответного токена достигает ROC-AUC до 0.83 для предсказания неправильных ответов в доменах, зависящих от знаний, тогда как подходы Model-as-Judge работают почти случайно. Во-вторых, мы исследуем геометрическую структуру мультиязычных представлений в латентном пространстве LLM, обнаруживая, что языковые направления могут быть идентифицированы с помощью PCA с точностью классификации 96–100% для четырёх языковых пар. Проецируя компоненту исходного языка из скрытых состояний, мы уменьшаем дистрибуционную дивергенцию до 55% и достигаем снижения индекса переключения кода на 63–95% в сгенерированном тексте без дополнительного обучения. В-третьих, мы разрабатываем метод дообучения с учётом сложности, который использует энтропию для стратификации обучающих данных и применяет целевые интервенции: стандартное дообучение для обычных примеров и дистилляцию цепочки рассуждений для сложных. Этот подход повышает точность с 0.39 до 0.52 (Qwen-3B) и с 0.51 до 0.64 (Phi-4-Mini) на MMLU-Pro при использовании на 79–82% меньше обучающих данных, с обобщением результатов на датасеты GSM8K и MedMCQA.

В совокупности эти вклады демонстрируют, что доступные сигналы одного прямого прохода предоставляют ценную информацию для решения фундаментальных проблем внедрения LLM. Разработанные методы не требуют архитектурных модификаций или дорогостоящего переобучения, что делает их практичными для реальных приложений.

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Chapter 1
Introduction

Note on the use of AI tools: *In some portions of this document (20% - 30% of the entire text), Artificial Intelligence assistants, particularly Generative AI, have been used to improve, rephrase, and summarize the content. The technologies used include Claude Code.*

Large language models (LLMs) have emerged as one of the most transformative technologies in artificial intelligence, demonstrating remarkable capabilities across diverse domains including natural language understanding, code generation, mathematical reasoning, and creative writing [5, 37, 45]. These models, trained on vast corpora of text data, have shown an unprecedented ability to generalize across tasks and generate coherent, contextually appropriate responses. However, despite their impressive performance, the deployment of LLMs in real-world applications faces several fundamental challenges that limit their reliability, controllability, and efficiency.

Relevance. The widespread adoption of LLMs in critical applications—from healthcare diagnostics to legal document analysis, from educational tutoring to scientific research assistance—demands that these systems be trustworthy and predictable. Yet current LLMs exhibit three interconnected limitations that undermine their practical utility.

First, LLMs frequently produce incorrect or fabricated information with high apparent confidence, a phenomenon known as hallucination [22]. Users cannot easily distinguish between reliable and unreliable model outputs, making it difficult to trust LLM-generated content without extensive verification. The gap between a model's confidence and its actual correctness poses critical risks, particularly in high-stakes domains where overconfidence in incorrect answers leads to severe consequences.

Second, once trained, LLMs exhibit fixed behavioral patterns that may not align with user requirements. For multilingual models, this manifests as unwanted code-switching, where models unexpectedly switch languages mid-response [49]. More broadly, fine-grained control over generation behavior remains challenging without expensive retraining, leaving operators with limited options when a model's default behavior does not match deployment requirements.

Third, fine-tuning LLMs on new tasks or domains requires substantial computational resources. Current approaches often treat all training examples equally, ignoring the varying complexity and informativeness of different data points, leading to wasted computation on redundant or trivial examples. This inefficiency is particularly problematic for organizations with limited computational budgets that nonetheless need to adapt models to specialized domains.

These challenges are not merely academic concerns—they represent practical barriers to LLM adoption in high-stakes applications where errors carry significant consequences.

Main purpose of the research. This thesis investigates a unified approach to addressing these three challenges by leveraging signals that are readily available from a single forward pass through the model. Specifically, we exploit two types of signals: (1) *output distributions*, the probability distributions over tokens that models produce at each generation step, and (2) *internal representations*, the high-dimensional activation patterns in the model's hidden layers. Our central hypothesis is that these signals encode rich information about model behavior, uncertainty, and input characteristics that can be extracted and utilized without requiring architectural modifications or

additional training. Moreover, the interplay between these two signal types opens a path toward *uncertainty decomposition*: the geometric structure of internal representations can separate language-specific from language-agnostic components, potentially enabling more fine-grained analysis of why a model is uncertain.

The research addresses three concrete questions.

1. **RQ1**: How can patterns in token-level output distributions be used to characterize model uncertainty and predict answer correctness?

2. **RQ2**: What geometric structure exists in the latent representations of multilingual LLMs, and how can this structure be exploited to control language-specific behaviors?

3. **RQ3**: How can output distribution characteristics guide the selection and processing of training data to improve fine-tuning efficiency?

Scientific novelty. This thesis makes several novel contributions to the field of LLM research.

We present a systematic analysis of token-wise entropy as an uncertainty indicator, demonstrating that single-token answer entropy effectively predicts answer correctness in knowledge-dependent domains (ROC-AUC up to 0.83) while revealing that entropy primarily captures knowledge uncertainty rather than reasoning difficulty. This work was recognized with the **Best Paper Award at AINL 2025**.

We discover and characterize language-specific directions in LLM latent space that can be identified through PCA with 96–100% classification accuracy across four language pairs. We demonstrate that projecting out the source language component from hidden states reduces code-switching by 63–95% in generated text without any additional training, providing a new mechanism for behavioral control.

We develop a complexity-aware fine-tuning pipeline that combines entropy-based data stratification with targeted training interventions—direct SFT for regular examples and chain-of-thought distillation for hard examples—achieving accuracy improvements while reducing training data requirements by 79–82%, validated across three datasets and three model architectures.

Finally, we establish a unifying framework showing how signals from the forward pass, whether from output distributions or internal representations, can address diverse challenges in LLM deployment, suggesting broader applicability of these techniques.

Statements for defense.

1. Single-token answer entropy provides an effective signal for predicting LLM answer correctness in knowledge-dependent domains, achieving ROC-AUC scores up to 0.83 on the MMLU-Pro benchmark, while Model-as-Judge approaches perform near-randomly (ROC-AUC $\approx$ 0.49–0.57).

2. Language-specific directions exist in the latent space of multilingual LLMs and can be reliably identified through PCA with 96–100% classification accuracy across four language pairs. Projecting out the source language component from hidden states reduces code-switching by 63–95% as measured by Code-Switching Index in generated text.

3. Complexity-aware fine-tuning that stratifies training data by entropy and applies appropriate training strategies (direct SFT for regular examples, chain-of-thought distillation for hard ones) improves model accuracy while requiring 79–82% less training data than full distillation, generalizing across three datasets and three model architectures.

4. Signals from a single forward pass through an LLM—both output distributions and internal representations—encode sufficient information to address fundamental challenges in reliability, control, and training efficiency without requiring architectural modifications.

Chapter 2
Author contribution

This thesis is based on three research papers that have been submitted to peer-reviewed venues. Below we describe the author's specific contributions to each work.

Paper 1: When an LLM is Apprehensive About Its Answers — And When Its Uncertainty is Justified

Authors: P. Sychev, A. Goncharov, D. Vyazhev, E. Khalafyan, A. Zaytsev

Venue: Proceedings of AINL 2025 (**Best Paper Award**). arXiv preprint arXiv:2503.01688.

Summary: This paper investigates uncertainty estimation in large language models by analyzing token-level entropy patterns across different question topics and reasoning requirements. We demonstrate that entropy-based features can predict answer correctness with ROC-AUC up to 0.83 in knowledge-dependent domains, while Model-as-Judge (MASJ) performs near-randomly. We also reveal internal biases in the MMLU-Pro dataset regarding question complexity.

Author contribution: The thesis author (A. Goncharov) contributed to the experimental design, implementation of entropy analysis methods, data collection and processing, and analysis of results. The author participated in developing the methodology for extracting and analyzing token-wise entropy patterns and contributed to writing the experimental sections of the paper.

Paper 2: Complexity-Aware Fine-Tuning

Authors: A. Goncharov, D. Vyazhev, P. Sychev, E. Khalafyan, A. Zaytsev

Venue: Proceedings of EACL 2026. arXiv preprint arXiv:2506.21220.

Summary: This paper presents a complexity-aware fine-tuning pipeline that uses single-token answer entropy to stratify training data by complexity and applies targeted training interventions: standard SFT for regular examples and chain-of-thought distillation for hard examples. The approach achieves significant accuracy improvements while using 79–82% less training data than full distillation baselines, validated across three datasets (MMLU-Pro, GSM8K, MedMCQA).

Author contribution: The thesis author (A. Goncharov) is the lead author and primary contributor. The author designed the overall methodology, implemented the entropy-based data stratification system, developed the training pipeline combining SFT and chain-of-thought distillation, conducted all experiments, analyzed results, and wrote the majority of the paper.

Paper 3: Language Steering in Latent Space to Mitigate Unintended Code-Switching

Authors: A. Goncharov, N. Kondusov, A. Zaytsev

Venue: Proceedings of AINL 2026. arXiv preprint arXiv:2510.13849.

Summary: This paper explores the geometric structure of multilingual representations in LLM latent space. We discover that language-specific directions can be identified through

PCA with 96–100% classification accuracy across four language pairs (English–Chinese, English–Spanish, English–Russian, English–Hindi). Steering activations via projection onto these directions reduces distributional divergence by up to 55% and achieves 63–99% reduction in Code-Switching Index in generated text.

Author contribution: The thesis author (A. Goncharov) is the lead author and primary contributor. The author conceived the idea of using PCA to identify language directions, designed and implemented the steering methodology, conducted experiments across multiple model architectures and language pairs, performed the analysis of results, and wrote the majority of the paper.

Chapter 3
List of publications

1. Sychev P., Goncharov A., Vyazhev D., Khalafyan E., Zaytsev A. When an LLM is Apprehensive About Its Answers — And When Its Uncertainty is Justified // Proceedings of AINL 2025 (**Best Paper Award**). arXiv preprint arXiv:2503.01688.

2. Goncharov A., Vyazhev D., Sychev P., Khalafyan E., Zaytsev A. Complexity-Aware Fine-Tuning // Proceedings of EACL 2026. arXiv preprint arXiv:2506.21220.

3. Goncharov A., Kondusov N., Zaytsev A. Language Steering in Latent Space to Mitigate Unintended Code-Switching // Proceedings of AINL 2026. arXiv preprint arXiv:2510.13849.

Chapter 4
Literature review

This chapter surveys the research landscape relevant to the three main themes of this thesis: uncertainty estimation in language models, interpretability and control of internal representations, and efficient training methodologies. For each area, we review the current state of the art, identify key limitations, and position our contributions within the broader research context.

4.1 Uncertainty Estimation in Large Language Models

The problem of uncertainty quantification in neural networks has a long history [15], but the advent of large language models has introduced new challenges and opportunities. Traditional approaches to uncertainty estimation, such as Monte Carlo dropout [15] and deep ensembles [29], require multiple forward passes through the model and are computationally prohibitive for models with billions of parameters. A single forward pass through a 70B-parameter model already requires significant computational resources; running it dozens or hundreds of times for ensemble-based uncertainty estimation is impractical in real-time applications.

This computational constraint has motivated research into more efficient alternatives. Confidence calibration methods attempt to align model confidence scores with actual accuracy [16], but LLMs often exhibit poor calibration, particularly on out-of-distribution inputs [24]. The gap between a model's apparent confidence and its actual correctness—often called the confidence gap—poses critical risks in high-stakes applications where overconfidence in incorrect answers leads to severe consequences. In domains such as healthcare, law, education, and economics, an LLM's overestimation of its accuracy could result in harmful decisions or misinformation.

Verbalized confidence, where models are prompted to express uncertainty in natural language [25, 31], offers an interesting alternative but relies on the model's ability to accurately self-assess—a capability that is itself uncertain and may not generalize across domains. Token-level probability analysis has emerged as a more principled direction. Kuhn et al. [28] introduced semantic entropy, which clusters generated sequences by meaning before computing entropy, addressing the challenge of semantically equivalent but lexically different outputs. Malinin and Gales [33] proposed using the entropy of the predictive distribution as an uncertainty measure for sequence generation tasks. Fadeeva et al. [13] provided a comprehensive comparison of white-box and black-box uncertainty estimation methods, presenting promising results for entropy-based aggregation approaches, while their subsequent work [12] demonstrated the potential of token-level uncertainty for fact-checking LLM outputs.

However, existing methods tend to answer how good an overall confidence estimate is while ignoring why uncertainty appeared in a specific case and how it relates to the nature of the question. Furthermore, many existing methods generalize poorly for concise multiple-choice questions where the model's response consists of only one or two tokens. The relationship between entropy patterns during generation and answer correctness, particularly how this relationship varies across question topics and reasoning requirements, remains underexplored. Our research addresses this gap by systematically analyzing how single-token answer entropy correlates with model reliability across different domains and complexity levels.

4.2 Internal Representations and Interpretability

Understanding the internal representations of neural networks has been a central goal of interpretability research [36, 10]. For language models, this involves understanding what information is encoded in activation patterns and how this information is processed across layers.

Probing classifiers have been widely used to identify what linguistic and semantic information is encoded in model representations [2, 8]. These studies have revealed that different layers capture different types of information, with lower layers encoding syntactic features and higher layers encoding more semantic content [21]. This layered organization suggests that models develop increasingly abstract representations as information flows through the transformer stack.

More recently, researchers have discovered that model representations exhibit meaningful geometric structure beyond what probing classifiers capture. Linear directions in activation space have been found to correspond to interpretable concepts [34, 39], leading to the "linear representation hypothesis"—the idea that many high-level concepts are represented as directions in the model's activation space. This insight has enabled activation steering approaches, where model behavior can be modified by adding or removing vectors from intermediate activations [46, 30]. Zou et al. [55] introduced representation engineering, a systematic framework that uses PCA on contrastive prompt pairs to identify concept directions (e.g., honesty, harmlessness) and steers activations along them to control model behavior.

In parallel with steering methods, concept erasure approaches have been developed to remove specific information from representations entirely. Iterative null-space projection (INLP) [43] removes protected attributes by projecting representations onto the null space of a linear classifier trained to predict those attributes. LEACE [3] achieves the same goal in closed form, providing perfect linear concept erasure without iterative optimization. These methods demonstrate that specific properties encoded in representations can be surgically removed, a finding that is closely related to our approach of removing language identity via projection.

For multilingual models, the question of how different languages are represented has received significant attention. Work on mBERT and XLM-R revealed that language identity concentrates in a few principal components of hidden states, separable from semantics when analyzed on parallel texts [9, 6]. Yang et al. [53] demonstrated that removing these language-specific components produces more language-agnostic embeddings, improving cross-lingual transfer. Wendler et al. [48] extended these findings to decoder-only LLMs, revealing that modern autoregressive models maintain interpretable geometric structure in their representations, with latent language preferences that can be identified through simple linear analysis. While these prior works focused primarily on analysis and understanding, our work takes the next step by exploiting this geometric structure for active generation control, using language identity as a practical test case for inference-time behavioral modification.

4.3 Code-Switching in Multilingual Models

Code-switching refers to the alternation between languages within a single conversation or text. While natural in multilingual human communication, unintended code-switching in LLM outputs is generally undesirable as it reduces output quality and user experience [11]. In deployed systems, such behavior degrades user trust: a customer-service chatbot prompted in Spanish that switches to English mid-response creates confusion rather than value. Recent work has also identified code-switching as a potential red-teaming vector, as language switches can bypass safety filters designed for specific languages.

Prior approaches to controlling code-switching have primarily relied on training-time interventions. Language-specific fine-tuning can reduce code-switching but requires separate models

or adapters for each language [41], which is expensive to maintain and does not scale to the hundreds of languages that modern LLMs nominally support. Constrained decoding methods can force outputs to remain in a target language but may degrade generation quality [42] by restricting the model's vocabulary in ways that interfere with fluent expression.

The discovery that language identity is encoded as linear directions in latent space [48] opens new possibilities for inference-time control that avoid the costs of retraining. However, practical methods for exploiting this structure to reduce code-switching have not been thoroughly explored. Our work addresses this gap by developing and evaluating activation steering methods specifically designed for language control, demonstrating that a simple projection operation applied to the final layers can substantially reduce code-switching without any additional training.

4.4 Efficient Fine-Tuning and Data Selection

Fine-tuning large language models is computationally expensive, motivating research into more efficient approaches at multiple levels. Parameter-efficient fine-tuning methods such as LoRA [20] and adapters [18] reduce the number of trainable parameters, making fine-tuning feasible on consumer hardware. However, these methods do not address data efficiency—they still train on all available data regardless of its informativeness.

Data selection and curriculum learning aim to improve training efficiency by carefully choosing which examples to train on and in what order. Influence functions [27] can identify the most impactful training examples but are prohibitively expensive to compute for models with billions of parameters. Active learning approaches [44] select informative examples iteratively but require multiple training rounds, each potentially as expensive as full fine-tuning. The LIMA approach [54] demonstrated that as few as 1,000 carefully curated high-quality examples suffice for alignment, showing that data quality and curation can substitute for data volume—but manual curation does not scale.

Recent work has explored using model predictions to guide data selection automatically. Self-training and self-distillation methods [14, 51] use model outputs as targets for further training. Chain-of-thought distillation [17, 32, 19] transfers reasoning capabilities from larger to smaller models by training on generated rationales, achieving strong performance with smaller model sizes. The SmallToLarge approach [26] uses small models to summarize training trajectories and guide data selection for larger models, demonstrating that cheap proxy signals can inform expensive training decisions.

The idea of stratifying training data by difficulty has been explored in curriculum learning [4, 26], where models are trained on progressively harder examples. However, curriculum-style approaches have shown limited gains in final accuracy when applied to LLM fine-tuning, with extreme sensitivity to hyperparameters such as epoch counts and learning rates. Our work proposes a fundamentally different approach: rather than merely ordering examples by difficulty, we apply different training interventions to different complexity levels. Easy examples receive standard supervised fine-tuning, while hard examples receive chain-of-thought distillation. This insight—that different complexity levels benefit from different training strategies, not just different ordering—is the key contribution that distinguishes our approach from prior curriculum learning methods.

4.5 Summary and Research Gaps

The literature reveals several research gaps that this thesis addresses. While uncertainty estimation methods exist, the relationship between single-token answer entropy and answer correctness has not been systematically studied across different question domains and reasoning requirements, leaving practitioners without clear guidance on when to trust entropy-based confidence signals. Although

language-specific directions in latent space have been identified in prior analytical work, practical methods for using them to control code-switching at inference time—particularly methods that require no additional training—are lacking. Finally, existing data selection methods do not leverage the insight that examples of different complexity may benefit from fundamentally different training strategies rather than merely different ordering or weighting.

This thesis addresses these gaps through three interconnected studies that exploit signals from the model's forward pass to improve reliability, control, and efficiency.

Chapter 5
Problem statement

This chapter formally defines the research problems addressed in this thesis and establishes the notation used throughout.

5.1 Preliminaries and Notation

Let $\mathcal{M}$ denote an autoregressive language model with vocabulary $\mathcal{V}$ of size $|\mathcal{V}|$. Given an input sequence $x = (x_1, \ldots, x_n)$, the model produces a probability distribution over the next token:

$$p(x_{n+1}|x_1, \ldots, x_n) = \text{softmax}(f_\theta(x_1, \ldots, x_n)) \tag{5.1}$$

where $f_\theta : \mathcal{V}^n \to \mathbb{R}^{|\mathcal{V}|}$ represents the model's logit function parameterized by θ.

For a generated sequence $y = (y_1, \ldots, y_m)$, we define the token-wise entropy at position t as:

$$H_t = -\sum_{v \in \mathcal{V}} p(v|x, y_1, \ldots, y_{t-1}) \log p(v|x, y_1, \ldots, y_{t-1}) \tag{5.2}$$

Let $h_l^{(t)} \in \mathbb{R}^d$ denote the hidden state at layer l and position t, where d is the hidden dimension of the model.

5.2 Problem 1: Uncertainty Estimation via Output Distributions

Definition 5.1 (Answer Correctness Prediction) *Given a question q and a model-generated answer a, the answer correctness prediction problem is to estimate $P(\text{correct}|q, a, \mathcal{M})$ using only information available from a single forward pass through $\mathcal{M}$.*

The goal is to identify features derived from the output distribution that correlate with answer correctness. Specifically, we investigate the sequence of token-wise entropies $(H_1, H_2, \ldots, H_m)$ during answer generation, aggregate statistics derived from these sequences (including mean entropy, maximum entropy, minimum entropy, and their ratios), and the predictive power of these features for distinguishing correct from incorrect answers across different question domains and reasoning requirements.

Objective: Develop entropy-based features that achieve high ROC-AUC for predicting answer correctness, improving upon baseline confidence measures such as first-token probability, mean sequence probability, and verbalized confidence.

5.3 Problem 2: Language Control via Latent Space Steering

Definition 5.2 (Language Direction) *A language direction $v_\ell \in \mathbb{R}^d$ for language ℓ is a unit vector in the model's hidden state space such that the projection of hidden states onto v_ℓ is predictive of whether the model is generating text in language ℓ.*

Definition 5.3 (Code-Switching) *Code-switching occurs when a model, prompted to respond in language ℓ_1, produces tokens that belong to a different language $\ell_2 \neq \ell_1$.*

The language steering problem consists of two sub-problems:

Sub-problem 2a (Direction Identification): Given a multilingual model $\mathcal{M}$ and a set of languages $\mathcal{L}$, identify language directions $\{v_\ell\}_{\ell \in \mathcal{L}}$ from hidden states using unsupervised methods.

Sub-problem 2b (Activation Steering): Given identified language directions, modify the model's hidden states during inference to reduce code-switching by projecting out the source language component:

$$\tilde{h}_l^{(t)} = h_l^{(t)} - s \left(h_l^{(t)} \cdot v^{(l)} \right) v^{(l)} \tag{5.3}$$

where $s \in \mathbb{R}$ controls intervention strength. When $s = 1$, this is an exact orthogonal projection that removes the language component; values $|s| > 1$ actively push representations away from the source language direction.

Objective: Achieve high language classification accuracy ($> 95\%$) for direction identification and significant reduction in code-switching ($> 50\%$ reduction in KL divergence; $> 60\%$ reduction in Code-Switching Index in generated text).

5.4 Problem 3: Complexity-Aware Fine-Tuning

Definition 5.4 (Example Complexity) *The complexity of a training example (x, y) with respect to model $\mathcal{M}$ is measured by the entropy of the model's output distribution when generating y given x:*

$$C(x, y; \mathcal{M}) = \frac{1}{|y|} \sum_{t=1}^{|y|} H_t \tag{5.4}$$

The complexity-aware fine-tuning problem is to design a training procedure that stratifies training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ into complexity tiers based on $C(x_i, y_i; \mathcal{M})$, applies appropriate training interventions for each tier—direct supervised fine-tuning (SFT) for low-complexity examples and chain-of-thought distillation from a teacher model for high-complexity examples—and achieves higher accuracy with less training data compared to uniform training.

Formally, let $\mathcal{D}_{\text{easy}} = \{(x, y) \in \mathcal{D} : C(x, y; \mathcal{M}) < \tau\}$ and $\mathcal{D}_{\text{hard}} = \mathcal{D} \setminus \mathcal{D}_{\text{easy}}$ for some threshold τ. The training objective is:

$$\mathcal{L} = \sum_{(x,y) \in \mathcal{D}_{\text{easy}}} \mathcal{L}_{\text{SFT}}(x, y) + \sum_{(x,y) \in \mathcal{D}_{\text{hard}}} \mathcal{L}_{\text{distill}}(x, y, \mathcal{M}_{\text{teacher}}) \tag{5.5}$$

Objective: Achieve accuracy improvements over baseline SFT while using significantly less training data ($> 50\%$ reduction in data requirements).

5.5 Unifying Theme

All three problems share a common approach: extracting and utilizing information from a single forward pass through the model. This information takes two forms. The first is output distributions—the probability distributions over tokens that the model produces at each generation step—which are exploited in Problems 1 and 3 for uncertainty estimation and complexity-aware training respectively. The second is internal representations—the hidden states at intermediate layers of the transformer—which are exploited in Problem 2 for language direction identification and activation steering.

The unifying hypothesis is that these signals encode rich information about model behavior, uncertainty, and input characteristics that can be exploited without architectural modifications or additional training. This thesis validates this hypothesis through empirical investigation of all three problems.

Beyond this shared methodology, the three problems are connected by a deeper theme: uncertainty and its structure. Output entropy measures total uncertainty, latent-space directions decompose representations into interpretable subspaces (e.g., language-specific vs. language-agnostic), and complexity-aware training acts on uncertainty to allocate resources. Together, these suggest that combining output-distribution and representation-level signals could enable structured uncertainty decomposition—disentangling, for instance, whether a model's uncertainty stems from a language-related ambiguity or from a genuine knowledge gap.

Chapter 6
Methodology

This chapter describes the theoretical approaches, methods, and algorithms used in each of the three studies comprising this thesis. For each study, we present the formal methodology, the key design choices, and the evaluation protocol.

6.1 Study 1: Uncertainty Estimation via Token-Wise Entropy

Entropy Extraction

For each generated token y_t in a model's response, we extract the full probability distribution over the vocabulary and compute the Shannon entropy:

$$H_t = -\sum_{v \in \mathcal{V}} p_t(v) \log_2 p_t(v) \tag{6.1}$$

where $p_t(v) = p(v|x, y_1, \ldots, y_{t-1})$ is the probability assigned to token v at position t. This entropy measures the model's uncertainty at each generation step: low entropy indicates that the model concentrates probability mass on a small number of tokens (high confidence), while high entropy indicates a diffuse distribution across many candidates (low confidence).

Uncertainty Estimation Approaches

We consider two fundamentally different approaches to uncertainty estimation: intrinsic entropy-based methods that analyze the model's own output distribution, and extrinsic Model-as-Judge methods that use an external evaluator.

Single-token answer entropy. For MMLU-Pro questions with multiple-choice answers, the expected answer consists of one or two tokens (the option number). We prompt the model to answer directly without chain-of-thought reasoning and compute the Shannon entropy of the output probability distribution over the answer token. This serves as our primary complexity metric, based on the assumption that response uncertainty is a natural predictor of question complexity: questions the model finds easy produce low-entropy (confident) predictions, while difficult questions produce high-entropy (uncertain) predictions.

Chain-of-thought entropy variants. We also analyze entropy when the model produces a chain-of-thought response, hypothesizing that the reasoning process may incrementally accumulate entropy and provide better separation of certain and uncertain answers. From the CoT response, we compute several aggregation features. CoT Mean computes the average entropy across all tokens in the reasoning chain. CoT Max captures the maximum entropy, corresponding to the point of greatest uncertainty during reasoning. Sequence-aggregation methods compute mean or maximum entropy at the word level rather than the token level, accounting for subword tokenization. Finally, marginal difference measures the mean gap between the top-2 token probabilities, capturing how decisive the model is at each step.

Model-as-Judge (MASJ) Approach

As a complementary extrinsic approach, we use a large auxiliary LLM (Mistral-Large-Instruct-2411, 123B parameters) to estimate question properties along two dimensions. The reasoning requirement dimension assesses whether the question requires complex reasoning and estimates the number of logical steps needed (low, medium, or high). The education level dimension estimates the question difficulty as an educational level (high school and easier, undergraduate, graduate, or postgraduate). The MASJ scores are obtained via carefully designed prompts (provided in the Appendix) and filtered using an MT-Bench-style quality assessment: instances where the judge assigns a quality score below 8 on a scale of 1 to 10 are regenerated to ensure consistent evaluation quality.

Evaluation Protocol

We compare uncertainty estimates with ground-truth correctness labels using ROC-AUC, which measures the discriminative power of each signal for separating correct from incorrect answers. The analysis is stratified by subject domain (14 topics in MMLU-Pro), reasoning requirement (no reasoning, low, medium, high), and model architecture. Answers with incorrect formatting, which constitute less than 5% of questions across all models, are excluded from the analysis.

6.2 Study 2: Language Steering in Latent Space

Hidden State Extraction

For a multilingual model $\mathcal{M}$ and a parallel corpus $\mathcal{D} = \{(s_i, k_i)\}_{i=1}^{N}$ where s_i represents semantically equivalent content in language $k \in \mathcal{K}$, we extract hidden states $h_i^{(\ell)} \in \mathbb{R}^d$ from each layer ℓ of the LLM. Specifically, we use the hidden state at the last token position of each sentence, as this position aggregates information from the entire input sequence through the causal attention mechanism. We use the Flores Plus dataset [35] with 50 parallel samples across five languages (English, Spanish, Russian, Chinese, Hindi) for PCA fitting and 100 additional samples for validation. The parallel nature of the corpus is essential: because the semantic content is held constant across languages, any systematic variance in the hidden states must be attributable to language identity rather than meaning.

Language Direction Identification via PCA

We apply Principal Component Analysis (PCA) to the collected hidden states to identify language-specific directions. For each layer, we pool hidden states across all languages, compute the mean hidden state $\mu = \frac{1}{|H|}\sum_{h \in H} h$, and center the data by subtracting this mean. We then compute the principal components of the centered data. The first principal component v_1 typically captures the language direction, as language identity is the dominant source of variance when semantic content is controlled. Formally, the language direction v_{lang} satisfies:

$$v_{\text{lang}} = \arg\max_{\|v\|=1} \text{Var}(v^\top H) \tag{6.2}$$

To validate that the identified direction genuinely captures language information rather than some other property, we train a logistic regression classifier on the scalar projection of hidden states onto the first principal component:

$$\hat{\ell} = \text{sign}(v_{\text{lang}}^\top h + b) \tag{6.3}$$

where b is a learned bias term. Classification accuracy above 95% indicates successful language direction identification, confirming that language identity is linearly accessible from a single scalar projection.

Activation Steering

During inference, we modify the model's hidden states to reduce code-switching by projecting out the source language component. For each token's hidden state $h_l^{(t)}$ at layer l, we apply:

$$\tilde{h}_l^{(t)} = h_l^{(t)} - s \left(h_l^{(t)} \cdot v^{(l)} \right) v^{(l)} \tag{6.4}$$

The parameter $s \in \mathbb{R}$ controls intervention strength. When $s = 1$, this is an exact orthogonal projection that removes the language component entirely. Values $0 < s < 1$ partially attenuate the language signal, while $|s| > 1$ goes beyond projection removal, actively pushing representations away from the source language direction. In practice, optimal coefficients have $|s| > 1$, indicating that mere removal of the language component is insufficient to overcome the language bias present in code-switched input; active counter-steering is needed.

Steering is applied only to the final layers where language variance concentrates, as determined by the explained variance analysis described in the results chapter. For Qwen2.5-1.5B (28 layers), we steer layers 27–28; for Llama-3.2-1B (16 layers), we steer layers 15–16. The optimal steering coefficient s is selected via grid search over a range of values, optimizing either KL divergence between the steered model's output distribution and the target language baseline (distributional optimization) or Code-Switching Index in generated text (generation-based optimization).

Evaluation Metrics

We evaluate steering effectiveness using two complementary approaches. Distributional evaluation compares the next-token prediction distribution of the steered model against the English baseline using four metrics: KL divergence $\mathrm{KL}(P_{\text{EN}} \| P_{\text{steered}})$ where lower divergence indicates better mitigation; Jensen-Shannon divergence as a symmetric measure of distributional similarity; cosine similarity between the steered and English baseline distributions; and top-50 token overlap, measuring the fraction of shared high-probability tokens with the English baseline.

Generation-based evaluation complements the distributional analysis by measuring code-switching in actual generated text. We generate 100-token continuations and compute the Code-Switching Index (CSI), defined as the fraction of tokens not in the target language; the M-Index, a multilingual diversity measure ranging from 0 (monolingual) to 1 (maximally mixed); the I-Index, measuring the ratio of language switch points to token boundaries; and Target Language Consistency (TLC $= 1 -$ CSI). Token-level language identification is performed using a FastText classifier with a sliding window of 5 tokens.

6.3 Study 3: Complexity-Aware Fine-Tuning

Complexity Estimation

For each training example (x, y), we estimate complexity by running the base (student) model on x and measuring the entropy of the answer token's probability distribution:

$$C(x, y) = H(p(\cdot|x)) = -\sum_{v \in \mathcal{V}} p(v|x) \log p(v|x) \tag{6.5}$$

High complexity indicates that the model is uncertain about the correct response, suggesting the example is difficult for this particular model. This is the same single-token answer entropy signal used in Study 1, but applied here for a different purpose: rather than predicting correctness at inference time, we use it to guide training data selection.

Data Stratification

We partition the training data into three complexity tiers by sorting all examples by entropy in ascending order and dividing into quartiles. The bottom quartile (lowest entropy) forms the easy set, the middle two quartiles form the medium set, and the top quartile (highest entropy) forms the hard set:

$$\mathcal{D}_{\text{easy}} = \mathcal{D}[1 : \lfloor N/4 \rfloor] \tag{6.6}$$

$$\mathcal{D}_{\text{medium}} = \mathcal{D}[\lfloor N/4 \rfloor : \lfloor 3N/4 \rfloor] \tag{6.7}$$

$$\mathcal{D}_{\text{hard}} = \mathcal{D}[\lfloor 3N/4 \rfloor + 1 : N] \tag{6.8}$$

The easy and medium examples are combined into a "regular" group $\mathcal{D}_{\text{regular}} = \mathcal{D}_{\text{easy}} \cup \mathcal{D}_{\text{medium}}$ that receives standard supervised fine-tuning, while hard examples receive chain-of-thought distillation. This asymmetric treatment reflects the hypothesis that easy and medium questions can be learned through direct supervision, while hard questions require the additional reasoning scaffolding provided by chain-of-thought reasoning from a larger teacher model. Since the number of hard questions is smaller, we balance the groups by filtering medium and easy examples to match the hard group size, ensuring equal representation during training.

Training Interventions

Regular examples (easy and medium complexity) are trained with standard supervised fine-tuning using the cross-entropy loss:

$$\mathcal{L}_{\text{SFT}}(x, y) = -\sum_{t=1}^{|y|} \log p_\theta(y_t | x, y_{<t}) \tag{6.9}$$

Hard examples receive chain-of-thought distillation from a teacher model ensemble $\mathcal{M}_T$. First, reasoning chains r are generated from the teacher by prompting it to think step-by-step. Then the student model is trained on the augmented examples that concatenate the reasoning chain with the answer:

$$\mathcal{L}_{\text{distill}}(x, y) = -\sum_{t=1}^{|r|+|y|} \log p_\theta((r \oplus y)_t | x, (r \oplus y)_{<t}) \tag{6.10}$$

where $r \oplus y$ denotes concatenation of the reasoning chain and the answer. This forces the student model to learn not just the correct answer but also the reasoning process that leads to it.

Training Pipeline

Algorithm 1 presents the complete complexity-aware fine-tuning pipeline in pseudocode form.

The pipeline proceeds in five stages. First, the student model computes entropy scores for all training examples via a single forward pass. Second, examples are sorted by entropy and partitioned into complexity tiers. Third, the student is fine-tuned on regular (easy + medium) examples using standard SFT for E_1 epochs. Fourth, the teacher model generates chain-of-thought reasoning for hard examples. Fifth, the student is fine-tuned on the reasoning-enriched hard examples for E_2

Algorithm 1 Complexity-Aware Fine-Tuning Pipeline

Require: Training dataset $\mathcal{D} = \{(q_i, a_i)\}_{i=1}^{N}$, Student LLM $\mathcal{M}_S$, Teacher LLM $\mathcal{M}_T$
Ensure: Fine-tuned Student LLM $\mathcal{M}_S^*$

// Step 1: Uncertainty Estimation
for each $(q_i, a_i) \in \mathcal{D}$ **do**
  $\mathbf{p}_i \leftarrow \mathcal{M}_S(q_i)$ {Get token probability distribution}
  $h_i \leftarrow -\sum_{j=1}^{|\mathcal{V}|} p_{ij} \log p_{ij}$ {Compute answer token entropy}
end for
// Step 2: Data Aggregation by Complexity
Sort $\mathcal{D}$ by entropy values $\{h_i\}_{i=1}^{N}$ in ascending order
$\mathcal{D}_{\text{easy}} \leftarrow \mathcal{D}[1 : \lfloor N/4 \rfloor]$
$\mathcal{D}_{\text{medium}} \leftarrow \mathcal{D}[\lfloor N/4 \rfloor + 1 : \lfloor 3N/4 \rfloor]$
$\mathcal{D}_{\text{hard}} \leftarrow \mathcal{D}[\lfloor 3N/4 \rfloor + 1 : N]$
$\mathcal{D}_{\text{regular}} \leftarrow \mathcal{D}_{\text{easy}} \cup \mathcal{D}_{\text{medium}}$
// Step 3: Standard SFT on Regular Data
for epoch $= 1$ to E_1 **do**
  for batch $\mathcal{B} \subset \mathcal{D}_{\text{regular}}$ **do**
    $\mathcal{L}_{\text{SFT}} \leftarrow -\sum_{(q,a)\in\mathcal{B}} \log P_{\mathcal{M}_S}(a \mid q)$
    $\mathcal{M}_S \leftarrow \mathcal{M}_S - \eta \nabla \mathcal{L}_{\text{SFT}}$
  end for
end for
// Step 4: Distill Chain-of-Thought for Hard Data
for each $(q_i, a_i) \in \mathcal{D}_{\text{hard}}$ **do**
  $r_i \leftarrow \mathcal{M}_T(q_i)$ {Generate CoT reasoning from teacher}
  $\mathcal{D}_{\text{hard}}^{\text{CoT}} \leftarrow \mathcal{D}_{\text{hard}}^{\text{CoT}} \cup \{(q_i, r_i, a_i)\}$
end for
// Step 5: SFT with Reasoning on Hard Data
for epoch $= 1$ to E_2 **do**
  for batch $\mathcal{B} \subset \mathcal{D}_{\text{hard}}^{\text{CoT}}$ **do**
    $\mathcal{L}_{\text{CoT}} \leftarrow -\sum_{(q,r,a)\in\mathcal{B}} \log P_{\mathcal{M}_S}(r \oplus a \mid q)$
    $\mathcal{M}_S \leftarrow \mathcal{M}_S - \eta \nabla \mathcal{L}_{\text{CoT}}$
  end for
end for
return $\mathcal{M}_S^* \leftarrow \mathcal{M}_S$

epochs. In our experiments, we set $E_1 = E_2 = 5$ for 10-epoch runs and $E_1 = E_2 = 10$ for 20-epoch runs.

Hyperparameter Selection

The complexity metric is selected based on ROC-AUC comparison with alternatives including MASJ reasoning score, MASJ education level, cross-entropy, and various chain-of-thought entropy aggregations. Single-token answer entropy consistently provides the best discrimination between complexity levels. The teacher model ensemble consists of DeepSeek-V3-0324, Qwen-3-235B, and Llama-4-Maverick, chosen for their strong reasoning capabilities. All fine-tuning uses LoRA [20] with rank 64, alpha 128, and dropout 0.05, enabling parameter-efficient training. The optimizer is AdamW with a learning rate of 10^{-4} across all configurations. Full hyperparameter details are provided in the Appendix.

6.4 Implementation Details

All experiments are implemented in Python using the PyTorch framework [40] and the Hugging Face Transformers library [50]. Hidden state extraction and entropy computation are performed during standard forward passes without requiring gradient computation, adding negligible overhead to inference. For Study 2, activation steering is implemented using PyTorch forward hooks that modify hidden states in-place during generation, requiring only one dot product and one vector subtraction per token per steered layer. For Study 3, the complexity scoring step requires a single forward pass through the student model on the entire training set, which is a one-time cost amortized across all subsequent training epochs.

Chapter 7
Numerical experiments

This chapter presents the experimental setup and results for each of the three studies comprising this thesis. For each study, we describe the motivation, experimental configuration, main results, and their interpretation.

7.1 Study 1: Uncertainty Estimation via Token-Wise Entropy

The gap between a model's apparent confidence and its actual correctness poses critical risks in high-stakes applications. In domains such as healthcare, law, education, and economics, an LLM's overestimation of its own accuracy could result in harmful decisions or misinformation. Existing uncertainty estimation methods tend to evaluate how good an overall confidence estimate is, but often ignore why uncertainty appeared in a specific case and how it relates to the nature of the question being asked. Furthermore, many existing methods generalize poorly for concise multiple-choice questions where the model's response consists of only one or two tokens. This study investigates whether the entropy of the model's output distribution over the answer token can serve as a reliable indicator of answer correctness, and how this signal varies across question topics and reasoning requirements.

Experimental Setup

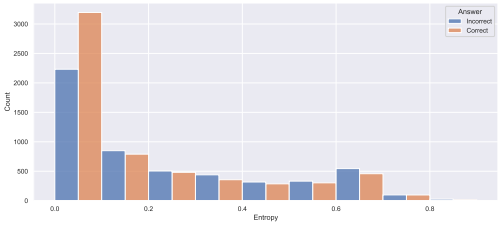
We use the MMLU-Pro benchmark [47], which contains approximately 12,000 challenging multiple-choice questions with 10 answer choices per question across 14 diverse domains including Biology, Chemistry, Computer Science, Economics, Engineering, Health, History, Law, Math, Philosophy, Physics, Psychology, Business, and Other. The dataset was chosen for its increased difficulty compared to the original MMLU and its improved robustness against data leakage, as official reports for Phi-4 and Mistral-Small-24B also used this extended dataset.

We evaluate four decoder-only large language models spanning a wide range of parameter counts: Phi-4 [1], Mistral-Small-24B-Base-2501 [23], Qwen-2.5-1.5B, and Qwen-2.5-72B [52]. This selection allows us to examine how model scale affects the relationship between entropy and answer correctness. For each model, we prompt it to answer questions directly (without chain-of-thought reasoning) and compute the Shannon entropy of the output probability distribution over the answer token. We measure the predictive power of this entropy signal using ROC-AUC for distinguishing correct from incorrect answers, stratified by subject domain and reasoning requirement.

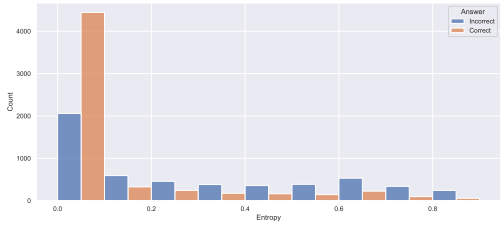
Entropy Distribution Analysis

Before examining predictive performance, we analyze the distribution of entropy values for correct and incorrect answers. Figure 7.1 presents entropy histograms for three models of different scales.

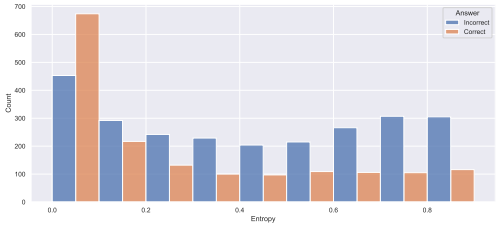
The distributions reveal a clear pattern: near-zero entropy values are substantially more common for correct answers across all models, confirming that when a model is highly confident (low entropy), it is more likely to be correct. However, the separation quality depends strongly on model scale. Larger models such as Qwen-72B exhibit clearer divergence between the entropy



(a) Phi-4



(b) Qwen-72B



(c) Qwen-1.5B

Figure 7.1: Entropy distribution of answer tokens for correct and incorrect responses. Near-zero entropy values are much more frequently observed for correct answers, particularly for larger models. Model scale correlates with entropy separation: larger models show clearer divergence between correct (low entropy) and incorrect (higher entropy) predictions.

distributions of correct and incorrect predictions, while smaller models like Qwen-1.5B show more overlapping distributions. This suggests that larger models develop better-calibrated internal representations of their own uncertainty.

Predictive Performance by Domain and Reasoning

Table 7.1 presents the ROC-AUC scores for predicting incorrect answers using single-token answer entropy across all 14 domains and reasoning complexity levels.

Figure 7.2 visualizes the ROC-AUC values across subjects for all four models, while Figure 7.3 shows the breakdown by reasoning requirement.

The results reveal several important patterns. Entropy is an effective predictor of answer correctness in knowledge-dependent domains, with Biology achieving the highest ROC-AUC of 0.83 for Qwen-72B, followed by Economics (0.80) and Computer Science (0.78). In contrast, performance is notably lower for reasoning-intensive domains: Math achieves only 0.55 ROC-AUC for Phi-4 and Law only 0.50. This domain dependence suggests that entropy primarily captures *knowledge uncertainty*—whether the model has encountered similar factual content during training—rather than *reasoning difficulty*.

The effect of model scale is substantial and consistent. Qwen-72B achieves an overall ROC-AUC of 0.77, compared to 0.70 for Qwen-1.5B (same architecture family but 48× fewer parameters), 0.58 for Phi-4, and 0.52 for Mistral. This scaling trend holds across all domains without exception, suggesting that larger models develop more informative entropy signals.

The reasoning analysis further supports the knowledge-versus-reasoning interpretation. For Qwen-72B, questions that do not require reasoning achieve ROC-AUC of 0.79, compared to 0.73 for questions requiring high reasoning. This gap is even more pronounced for Mistral (0.59 vs. 0.53). Low entropy is thus a marginally better predictor of correctness for factual recall questions than for multi-step reasoning problems.

The Model-as-Judge (MASJ) approach, in which a large auxiliary LLM estimates question

Table 7.1: ROC AUC performance comparison across categories and models using single-token answer entropy.

Category	Phi-4	Mistral	Qwen 1.5B	Qwen 72B
Biology	**0.73**	**0.65**	**0.79**	**0.83**
Economics	0.65	0.58	0.73	0.80
Philosophy	0.65	0.49	0.61	0.74
Physics	0.63	0.44	0.64	0.74
Computer Science	0.61	0.56	0.69	0.78
Psychology	0.61	0.65	0.77	0.77
Chemistry	0.61	0.44	0.64	0.70
Health	0.60	0.55	0.67	0.77
Business	0.58	0.52	0.64	0.74
History	0.58	0.58	0.65	0.67
Engineering	0.58	0.44	0.64	0.73
Other	0.56	0.53	0.66	0.78
Math	0.55	0.54	0.63	0.73
Law	0.50	0.57	0.58	0.69
No Reasoning	**0.59**	**0.59**	**0.72**	**0.79**
Yes Reasoning	0.56	0.52	0.68	0.76
Reasoning (Low)	0.59	0.58	0.71	0.79
Reasoning (Med)	0.57	0.52	0.69	0.76
Reasoning (High)	0.53	0.53	0.61	0.73
Overall	0.58	0.52	0.70	0.77

difficulty, performs near-randomly with ROC-AUC values around 0.49–0.57 across all models. Both the numerical and nominal MASJ scores relate little to actual question complexity, indicating that MASJ captures different aspects of a question's properties that are largely irrelevant to predicting model correctness.

Calibration Analysis

Figure 7.4 presents calibration curves computed using inverted normalized entropy. The analysis reveals systematic deviations between self-reported certainty and empirical accuracy, with distinct patterns across architectures.

Mistral exhibits unstable calibration dynamics, characterized by erratic accuracy fluctuations in low-confidence regions. Despite moderate confidence estimates, its accuracy varies unpredictably, even peaking anomalously in the lowest entropy bin. This contrasts with its pronounced underperformance in high-confidence regimes, where empirical accuracy consistently trails reported certainty. Qwen-72B, despite its large scale, demonstrates inconsistent calibration in high-confidence predictions, with substantial confidence-accuracy gaps in mid-range entropy bins. Interestingly, Qwen-1.5B emerges as an outlier, surpassing its larger counterpart in calibration quality: its confidence-accuracy relationship progresses monotonically across entropy bins with minimal extreme deviations. These results underscore that systematic overconfidence is a

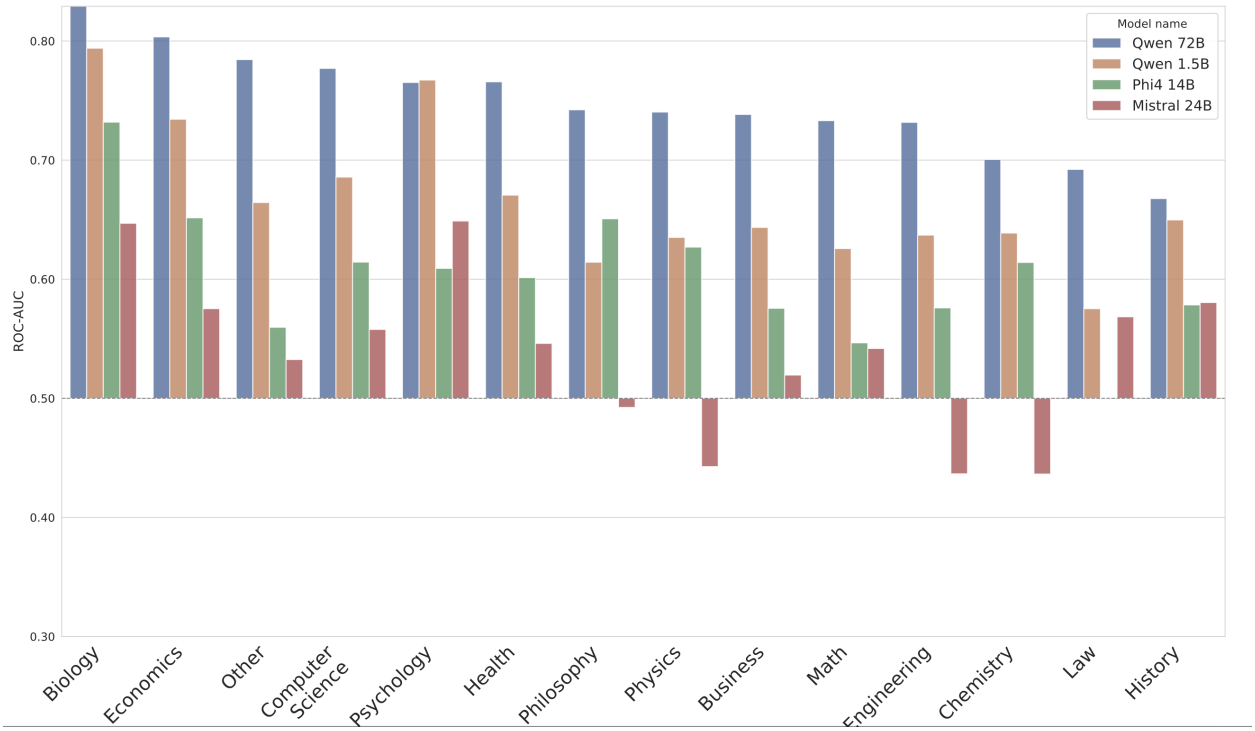


Figure 7.2: ROC-AUC for error prediction by subject for four different LLMs. Knowledge-intensive domains (Biology, Psychology) consistently show higher predictive power than reasoning-heavy domains (Math, Engineering).

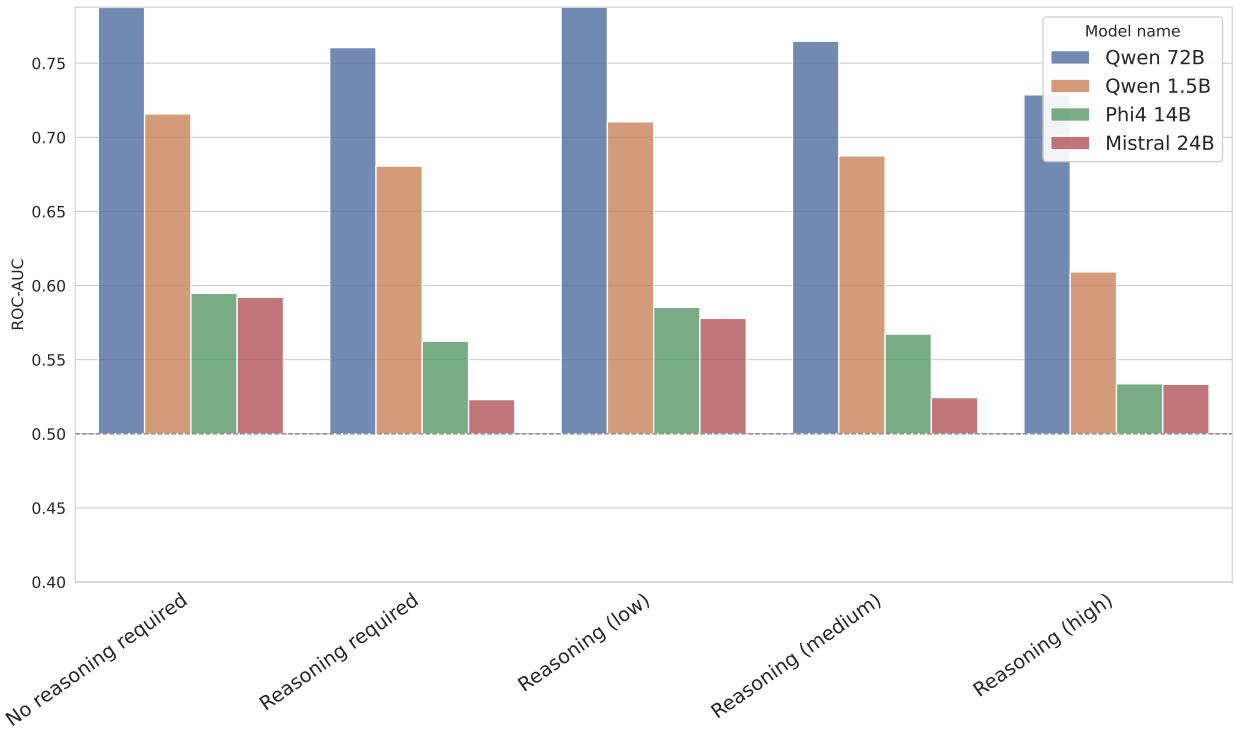


Figure 7.3: ROC-AUC by reasoning requirement for four different LLMs. Entropy is a better predictor for questions that do not require complex reasoning.

pervasive issue across model architectures, particularly in high-certainty regimes where all models significantly overestimate their reliability.

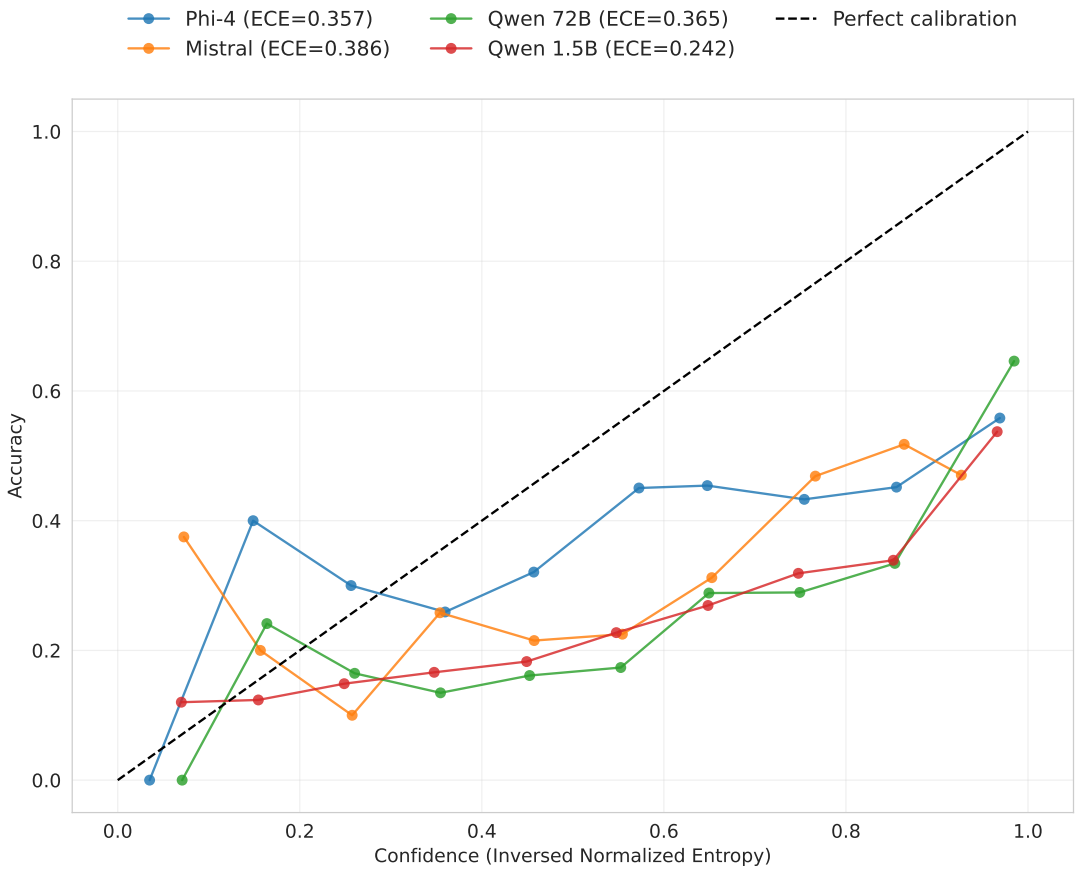


Figure 7.4: Calibration of LLMs based on their entropy. Systematic overconfidence is observed in high-certainty regimes across all models, with architecture-specific patterns.

Discussion

This study demonstrates that single-token answer entropy is a practical and effective signal for predicting LLM answer correctness, but with important caveats. The signal is most informative for knowledge-dependent domains where the model either knows the answer (low entropy) or does not (high entropy). For questions requiring multi-step reasoning, the entropy of the final answer token is less informative because the model may be uncertain about the reasoning process while still arriving at the correct answer, or conversely, may be confident in an incorrect reasoning chain.

The observation that MASJ performs near-randomly is significant: it suggests that external judges, even when using large auxiliary models, capture fundamentally different properties than the model's own internal uncertainty. This finding motivates the use of intrinsic signals (output distributions) rather than external evaluation for uncertainty estimation.

We also note that the MMLU-Pro dataset exhibits internal biases in question complexity. The distribution of reasoning requirements varies significantly by topic—engineering questions require reasoning in approximately 90% of cases, while philosophy requires it in only about 50%. This suggests that more balanced and less biased datasets would be needed for fair measurement of LLM progress across domains.

7.2 Study 2: Language Steering in Latent Space

Multilingual large language models frequently exhibit unintended code-switching—generating tokens in languages other than the target despite explicit monolingual instructions. In deployed sys-

tems, such behavior degrades user trust and experience. For instance, a customer-service chatbot prompted in Spanish may switch to English mid-response, confusing users. Existing solutions require costly fine-tuning on monolingual data, reinforcement learning with language-specific rewards, or brittle prompt engineering. This study investigates whether the geometric structure of multilingual representations in the model's latent space can be exploited for a cheaper alternative: inference-time activation steering that requires no additional training.

Experimental Setup

We evaluate two multilingual models of different architectures: Qwen2.5-1.5B with 28 layers and Llama-3.2-1B with 16 layers. We test four language pairs spanning different typological distances: English–Chinese (distant, different script), English–Spanish (close, same script), English–Russian (moderate, different script), and English–Hindi (distant, different script). This selection allows us to examine how typological similarity and script differences affect steering effectiveness.

For PCA fitting, we use the Flores Plus dataset [35] with 50 parallel samples per language and 100 samples for validation. The parallel nature of the corpus is essential: because the semantic content is held constant across languages, language identity becomes the dominant variance source in the hidden state space. For steering evaluation, we use 2,467 TED Talks samples with artificial code-switching, where each sample is truncated to 20 words, split at the midpoint, and the second half is translated to the target language using Gemini 2.5 Flash, producing texts that begin in English and switch mid-sentence.

Language Direction Identification

Table 7.2 shows the classification accuracy for identifying language from hidden state projections onto the first principal component at the final layer.

Table 7.2: Last layer language classifier accuracy using PCA-identified directions.

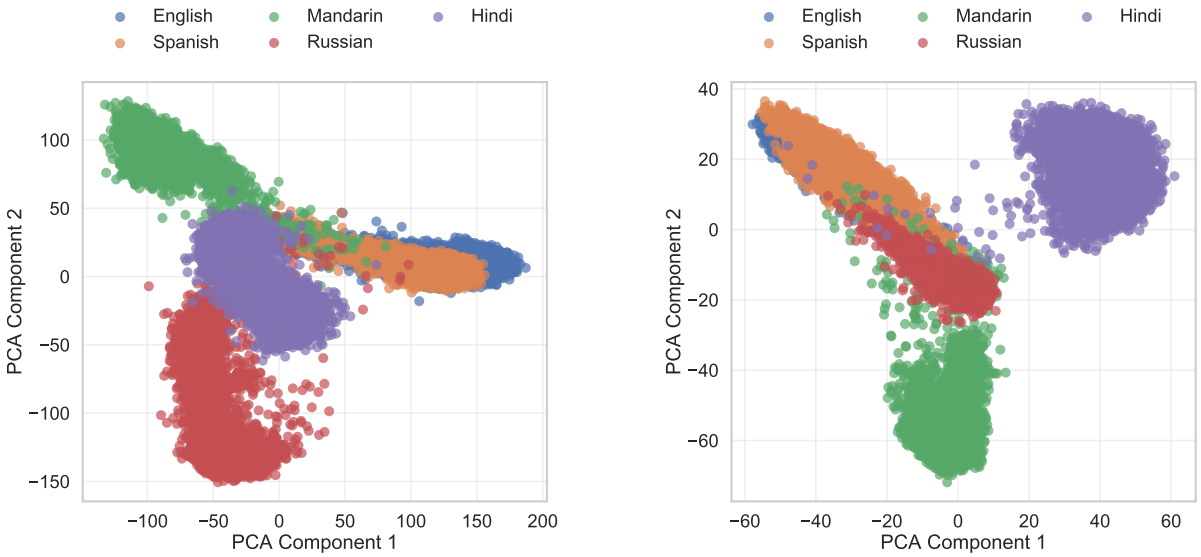
Language pair	Qwen	Llama
English – Chinese	0.98	0.98
English – Spanish	0.96	0.96
English – Russian	0.99	0.99
English – Hindi	0.98	1.00

A simple logistic regression classifier trained only on the scalar projection onto the first principal component achieves near-perfect accuracy (0.96–1.00) across all language pairs and both models. This validates that language identity information is linearly accessible in the final-layer representations, supporting the hypothesis that models maintain explicit language directions in their activation space. The slightly lower accuracy for English–Spanish (0.96 in both models) is attributable to these languages being more closely related within the Indo-European family, resulting in less separated clusters.

Layer-Wise Evolution of Language Structure

Figure 7.5 shows the final-layer PCA language maps for both models, while Figure 7.6 illustrates the layer-wise evolution of language clustering.

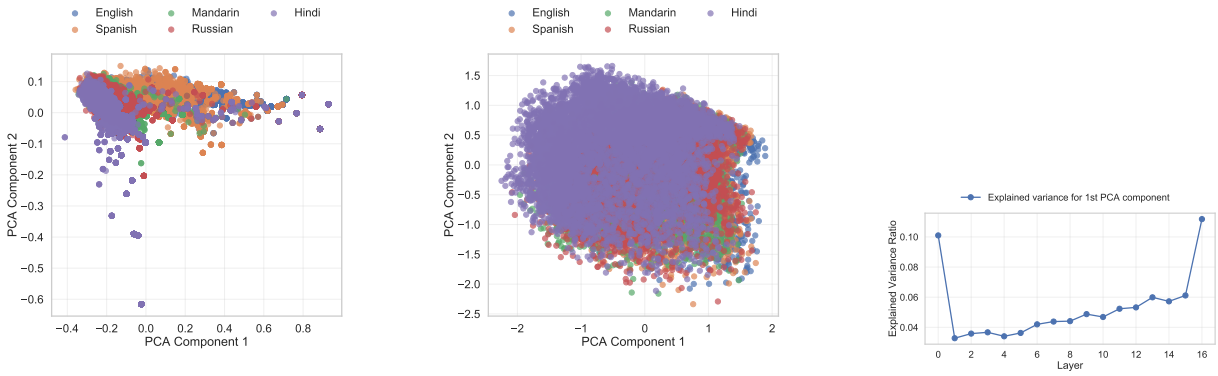
The visualizations reveal that models organize identical semantic content into tight, language-specific clusters with clear boundaries. Languages within the Indo-European family (Spanish, Russian) cluster closer together than typologically distant pairs (English–Chinese), suggesting that the



(a) Final layer language map (Qwen)

(b) Final layer language map (Llama)

Figure 7.5: Final-layer embeddings projected onto the first two principal components. Languages organize into tight, language-specific clusters with clear boundaries. Languages within the Indo-European family (Spanish, Russian) cluster closer than typologically distant pairs (English–Chinese).



(a) Layer 0 (Llama)

(b) Layer 7 (Llama)

(c) Explained variance by layer

Figure 7.6: Layer-wise evolution of language representations in Llama-3.2-1B. Clusters emerge loosely in early layers (a), sharpen in middle layers (b), and reach near-perfect separability in final layers. The explained variance of the first principal component (c) increases dramatically in final layers, confirming late-stage specialization for language identity.

first principal component captures coarse language distinction while the second component encodes finer linguistic properties. The layer-wise evolution is particularly informative: clusters emerge loosely in early layers and sharpen dramatically in final layers, with the explained variance of the first principal component increasing substantially in the last few layers. This confirms late-stage specialization for language identity, suggesting that early layers focus primarily on semantic processing while language identity crystallizes only in the final layers.

Distributional Evaluation of Steering

We evaluate steering effectiveness by comparing the next-token prediction distribution of the steered model against the English baseline using multiple complementary metrics.

Table 7.3: KL divergence reduction through activation steering (no steering → steering). Lower is better.

Language pair	Qwen	Llama
English – Chinese	10.15 → 4.52	9.05 → 4.77
English – Spanish	7.25 → 4.90	6.52 → 3.48
English – Russian	8.70 → 5.40	7.77 → 3.49
English – Hindi	9.60 → 7.02	8.06 → 6.13

Table 7.4: Cosine similarity between steered and English baseline distributions (no steering → steering, higher is better).

Language pair	Qwen	Llama
English – Chinese	0.02 → 0.22	0.05 → 0.21
English – Spanish	0.12 → 0.18	0.12 → 0.29
English – Russian	0.09 → 0.23	0.10 → 0.31
English – Hindi	0.04 → 0.07	0.06 → 0.16

Table 7.5: Best steering strength coefficient s (KL-optimized via grid search).

Language pair	Qwen	Llama
English – Chinese	-3.2	-5.0
English – Spanish	-2.8	-3.7
English – Russian	-2.2	-3.5
English – Hindi	$+5.0$	-3.6

Steering reduces KL divergence by 25–55% across language pairs and models, with an average reduction of 38% for Qwen and 43% for Llama. The method shows strongest performance for typologically distant pairs: English–Chinese achieves 55% KL reduction on Qwen, while English–Russian achieves 55% on Llama. We hypothesize that model-specific tokenization and training data balance play a key role in these differences: the Qwen model processed substantially more English and Chinese data during pretraining, leading to more granular tokenization and better internal generalization for that pair.

English–Hindi remains the weakest pair for both models, achieving only 24–27% KL reduction. The cosine similarity improvement for this pair is also minimal (0.04 → 0.07 for Qwen), reinforcing that this is the most challenging configuration. Notably, the optimal steering coefficient for Qwen English–Hindi is positive ($+5.0$), while all other pairs use negative coefficients. This indicates a reversed PCA direction for this pair, possibly reflecting tokenization asymmetries in the Qwen model's treatment of Devanagari script. In contrast, Llama uses a consistent negative direction across all language pairs.

The fact that optimal coefficients have $|s| > 1$ in all cases is significant: it indicates that merely removing the language component (which corresponds to $s = 1$) is insufficient. Active counter-steering beyond projection removal is needed to overcome the language bias present in code-switched input.

Generation-Based Evaluation

To complement the distributional analysis, we evaluate steering on actual text generation using Llama-3.2-1B. For each of 500 TED Talks samples per language pair, we generate 100-token continuations in three conditions: English-only input without steering, code-switched input without steering, and code-switched input with steering. Table 7.6 presents the results. All steered vs. unsteered differences are significant at $p < 0.001$ (Wilcoxon signed-rank test).

Table 7.6: Generation-based code-switching metrics (Llama-3.2-1B, $n = 500$). English-only baseline: CSI $= 0.02$.

	KL-optimized		CSI-optimized	
Language pair	CSI (↓)	TLC (↑)	CSI (↓)	TLC (↑)
Unsteered (code-switched input, no steering):				
English – Chinese	0.64 / 0.36		—	
English – Spanish	0.62 / 0.38		—	
English – Russian	0.59 / 0.41		—	
English – Hindi	0.81 / 0.19		—	
Steered:				
English – Chinese	0.03	0.97	0.03	0.97
English – Spanish	0.23	0.77	0.11	0.89
English – Russian	0.02	0.98	0.01	0.99
English – Hindi	0.04	0.96	0.01	0.99

With KL-optimized coefficients, the Code-Switching Index drops by 63–95% across language pairs. Target Language Consistency exceeds 0.77 for all pairs, with English–Chinese and English–Russian achieving near-perfect consistency (TLC > 0.97). CSI-optimized coefficients yield further improvements for English–Spanish (CSI from 0.23 to 0.11) and English–Hindi (CSI from 0.04 to 0.01), but at the cost of higher-magnitude coefficients that increase degeneration risk.

Discussion

The generation-based evaluation confirms that distributional optimization transfers well to actual text generation, but also reveals important fragility. Steering coefficients are highly sensitive to magnitude: small changes can shift the model from fluent code-switched output to degenerate repetition. The CSI-optimized coefficients cluster near the maximum tested value (-5.0), suggesting that direct optimization of generation metrics pushes toward an aggressive regime where language control and fluency trade off sharply. Qualitative inspection of generated text confirms this: for English–Spanish, steering successfully redirects generation from Spanish tokens to coherent English, but for English–Russian with aggressive coefficients, steering eliminates Russian but produces repetitive output patterns (e.g., "first year of the first year of the first year...").

This fluency–control trade-off represents a fundamental challenge for practical deployment. While the method achieves strong language consistency, the optimal operating point requires careful calibration per language pair, and no principled method for automatic coefficient selection has been developed. Combining steering with constrained decoding or adaptive coefficient scheduling represents a promising direction for addressing this limitation.

7.3 Study 3: Complexity-Aware Fine-Tuning

General-purpose LLMs excel across diverse tasks, but deploying them often requires domain-specific fine-tuning. A standard approach is to train on all available in-domain data with uniform supervised fine-tuning (SFT). However, recent work has concentrated primarily on optimizing training mechanics—objectives, schedulers, and regularizers—while comparatively little attention has been paid to data curation. Curriculum-style approaches have been explored with increasing complexity of data during training, but with limited gains in final accuracy and efficiency, suggesting that more principled strategies are needed. This study investigates whether entropy-based complexity estimation can guide a more efficient fine-tuning pipeline that applies different training strategies to examples of different difficulty.

Experimental Setup

We conduct experiments across three datasets that cover different domains and question types. MMLU-Pro provides approximately 12,000 multiple-choice questions across 14 domains with varying complexity. GSM8K [7] contains 8,500 grade school math word problems requiring multi-step arithmetic reasoning. MedMCQA [38] offers large-scale medical entrance exam questions covering 2,400 healthcare topics across 21 medical subjects.

For student models, we use three architectures: Qwen2.5-3B, Phi-4-Mini, and LLama-3.2-3B. For chain-of-thought distillation on hard examples, we use an ensemble of teacher models: DeepSeek-V3-0324, Qwen-3-235B, and Llama-4-Maverick. The data is randomly split into train, validation, and test sets with a 70:10:20 ratio, and balanced so that the number of entries in each complexity group is equal.

We compare our pipeline against four baselines. Standard SFT trains on all data without any complexity split. Curriculum-based SFT applies progressive difficulty ordering (3 epochs on easy data, 3 on medium, 4 on hard). Full distillation applies chain-of-thought distillation to all examples regardless of difficulty. The alternative pipeline reverses our approach: SFT on hard data first, then CoT distillation on easy and medium data.

Main Results

Table 7.7 presents the results after 10 epochs on MMLU-Pro, including both accuracy and total tokens processed to quantify data efficiency.

Table 7.7: Accuracy / tokens processed for complexity-aware fine-tuning pipelines after 10 epochs (MMLU-Pro).

Method	Qwen 3B	Phi4-mini
SFT	0.42 / 29k	0.55 / 27k
Curriculum	0.45 / 29k	0.54 / 27k
Distillation	<u>0.49</u> / 1.97m	**0.63** / 1.51m
Alternative	0.47 / 1.5m	0.58 / 1.2m
Pipeline (ours)	**0.50** / **399k**	<u>0.60</u> / **268k**

After 10 epochs, the complexity-aware pipeline achieves accuracy comparable to or exceeding full distillation while processing dramatically fewer tokens. Qwen-3B reaches 0.50 accuracy compared to 0.49 for distillation, but uses only 399k tokens versus 1.97 million—a reduction of 79%. Phi-4-Mini reaches 0.60 compared to 0.63 for distillation, using 268k tokens versus 1.51 million—a reduction of 82%. The pipeline also substantially outperforms the alternative ordering

(SFT on hard, then CoT on easy/medium), confirming that the direction of intervention matters: easy examples benefit from simple SFT, while hard examples require the additional reasoning scaffolding provided by chain-of-thought distillation.

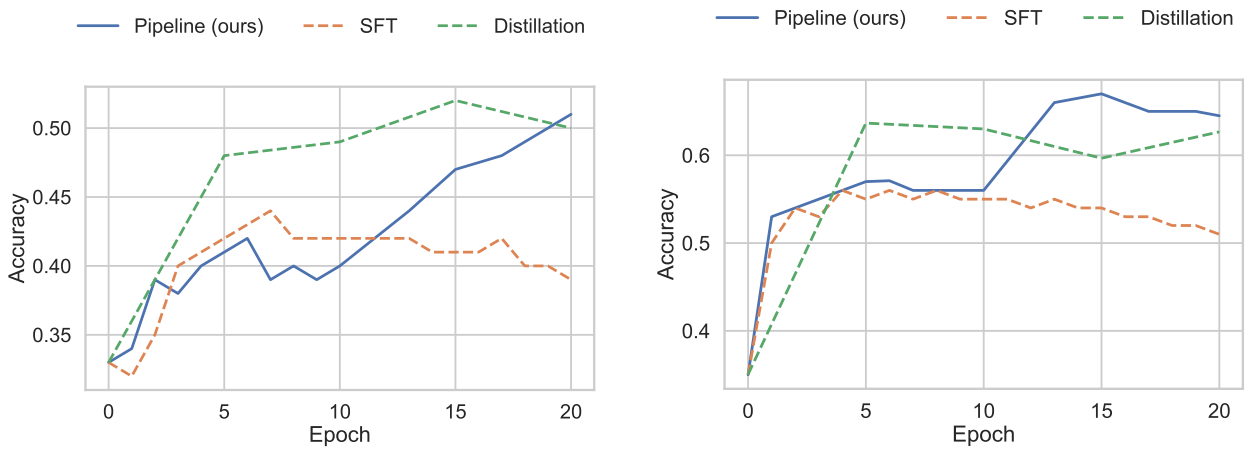
To examine longer-term training dynamics, we extended experiments to 20 epochs. Table 7.8 shows the results, including generalization to GSM8K and MedMCQA.

Table 7.8: Accuracy for complexity-aware fine-tuning pipelines after 20 epochs.

	MMLU Pro		GSM8K		MedMCQA	
Method	Qwen 3B	Phi4-mini	LLama 3B	Phi4-mini	LLama 3B	Phi4-mini
SFT	0.39	0.51	0.13	0.19	0.51	0.51
Distillation	0.50	0.63	0.79	0.89	0.56	0.57
Pipeline (ours)	**0.52**	**0.64**	**0.82**	**0.89**	**0.56**	**0.58**

After 20 epochs, the pipeline surpasses all baselines across all three datasets. On MMLU-Pro, Qwen-3B achieves 0.52 compared to 0.50 for full distillation and 0.39 for SFT, while Phi-4-Mini reaches 0.64 compared to 0.63 and 0.51 respectively. The method generalizes strongly to other domains: on GSM8K, LLama-3B achieves 0.82 (pipeline) versus 0.79 (distillation) and 0.13 (SFT), demonstrating particularly large gains for mathematical reasoning. On MedMCQA, the pipeline matches or slightly exceeds distillation performance while maintaining the same 79–82% data savings.

Figure 7.7 shows the training accuracy dynamics over 20 epochs, revealing that the pipeline exhibits a continued uptrend while full distillation plateaus after approximately 10 epochs. This suggests that the complexity-aware approach makes better use of training data by avoiding redundant computation on examples the model can already handle.



(a) Qwen 3B (MMLU-Pro, 20 epochs)

(b) Phi-4-Mini (MMLU-Pro, 20 epochs)

Figure 7.7: Accuracy dynamics for fine-tuning pipelines over 20 epochs. The complexity-aware pipeline (blue solid) surpasses full distillation (green dashed) and continues improving, while distillation plateaus.

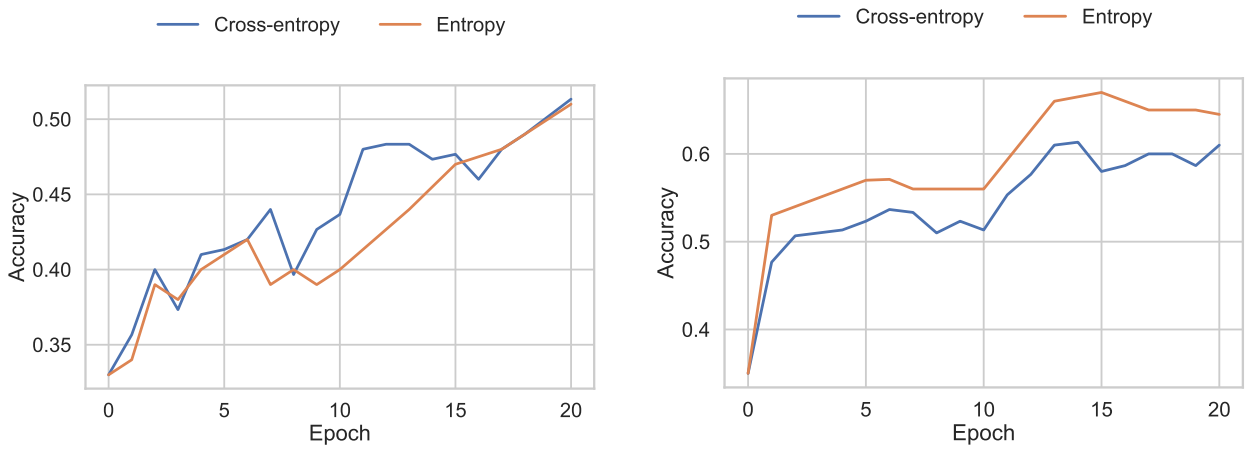
Sensitivity Study

Complexity Metric Selection

We compare single-token answer entropy against alternative complexity metrics: MASJ reasoning score, MASJ education level, cross-entropy, and various chain-of-thought entropy aggregations.

Single-token entropy yields the best ROC-AUC for complexity estimation (0.72 for Qwen-3B without domain split) and provides the clearest separation between easy and hard examples during SFT training. MASJ-based metrics achieve ROC-AUC around 0.50–0.57, barely above random, making them unsuitable for data stratification. Chain-of-thought entropy variants provide higher answer accuracy but lower ROC-AUC scores, making them less suitable for complexity estimation. Maximum-based measures (CoT Max, Sequence Mean Max) consistently outperform mean-based approaches, suggesting that peak uncertainty moments better indicate question difficulty than average uncertainty.

We also evaluate cross-entropy as an alternative to entropy for the pipeline's complexity metric. Figure 7.8 shows that entropy and cross-entropy achieve comparable pipeline performance for Qwen-3B, while Phi-4-Mini demonstrates superior performance with entropy. The difference can be attributed to entropy capturing information across the entire output distribution rather than focusing only on the correct token.



(a) Qwen 3B

(b) Phi-4-Mini

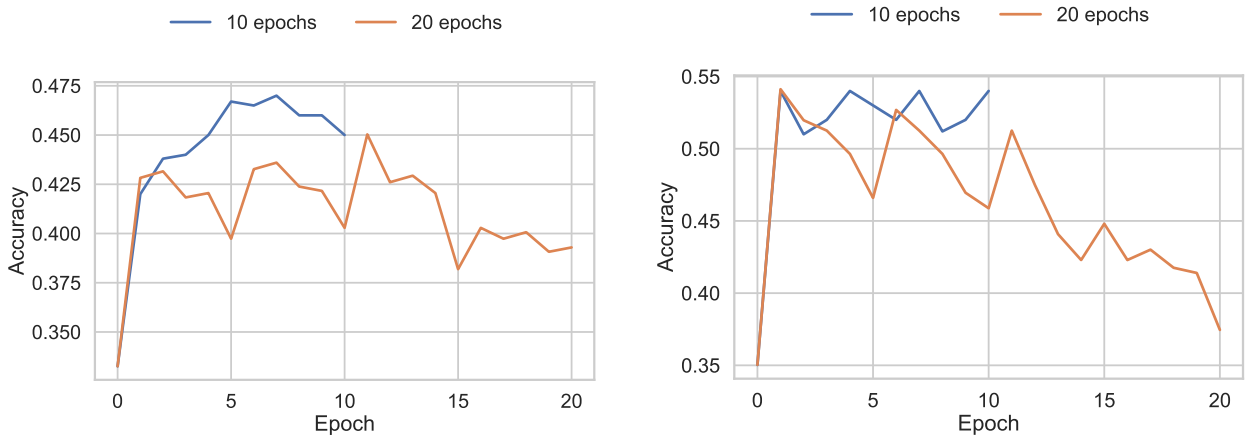
Figure 7.8: Pipeline performance comparison using entropy versus cross-entropy as the complexity metric. Entropy provides comparable or superior performance.

Curriculum Learning Fragility

Standard curriculum learning, which applies SFT with progressively increasing difficulty, shows extreme sensitivity to hyperparameters. Figure 7.9 illustrates this fragility: with 10 epochs of training, curriculum-based SFT achieves moderate performance, but extending to 20 epochs causes Qwen-3B to plateau at a lower accuracy while Phi-4-Mini quickly overfits and its performance degrades. This demonstrates that naive complexity-aware ordering without differentiated training strategies—using only SFT throughout—is insufficient. The key insight of our approach is that different complexity levels require fundamentally different training interventions, not merely different ordering.

Discussion

The results confirm that entropy works as a reliable, automatic proxy for question difficulty. The pipeline achieves accuracies of 0.52 and 0.64 compared to 0.39 and 0.51 for baseline SFT on Qwen-3B and Phi-4-Mini respectively, while using 81% less data than full distillation. This finding suggests that curriculum ideas remain relevant for modern LLMs: letting the model focus on what it can already solve directly, while providing extra reasoning guidance only where it struggles, yields a better allocation of limited model capacity.



(a) Qwen 3B

(b) Phi-4-Mini

Figure 7.9: Curriculum learning accuracy dynamics for different epoch configurations. Curriculum-based SFT shows extreme sensitivity to training duration, with Phi-4-Mini overfitting rapidly under extended training.

The pipeline is fully automated and can be integrated into existing fine-tuning workflows. All that is needed is a single forward pass through the student model to compute entropy scores, followed by sorting and partitioning the data. The distillation step for hard examples requires teacher model inference, but this applies to only approximately 25% of the training data, substantially reducing the cost compared to full distillation.

However, several limitations should be noted. Low entropy can still correspond to hallucinations, leading to imperfect complexity identification. The three-way equal split was chosen for simplicity but may not be optimal for all datasets. In low-resource settings where a teacher model is unavailable or imperfect, the benefit of chain-of-thought distillation is reduced. Finally, the approach has been validated only on question-answering tasks with clear correct answers; its applicability to open-ended generation remains to be investigated.

7.4 Summary of Results

Across all three studies, we demonstrate that signals from the model's forward pass provide valuable and complementary information for addressing key challenges in LLM deployment. Token-wise entropy achieves ROC-AUC up to 0.83 for predicting answer correctness in knowledge-dependent domains, with the signal quality improving with model scale, while the Model-as-Judge approach performs near-randomly. PCA-identified language directions enable 25–55% reduction in distributional divergence and 63–95% reduction in Code-Switching Index in generated text across four language pairs, using only simple linear projections applied to the final layers. Complexity-aware fine-tuning matches or outperforms full distillation while using 79–82% less training data, generalizing across three datasets and three model architectures, and demonstrating that automatic complexity estimation via entropy enables intelligent allocation of training resources.

These results validate our central hypothesis that accessible signals from forward passes—both output distributions and internal representations—encode sufficient information to improve reliability, control, and efficiency without requiring architectural modifications. The three studies are unified by the principle that a single forward pass contains rich information about the model's confidence, the input's characteristics, and the relationship between the two.

Chapter 8
Discussion and conclusion

8.1 Summary of Contributions

This thesis investigated how signals from a single forward pass through large language models—both output distributions and internal representations—can be leveraged to address fundamental challenges in LLM deployment. We presented three interconnected studies that demonstrate the practical utility of this approach.

In the first study, we demonstrated that single-token answer entropy provides an effective and practical signal for predicting answer correctness in knowledge-dependent domains. Across four models spanning parameter counts from 1.5B to 72B, we observed ROC-AUC scores up to 0.83 on the MMLU-Pro benchmark, with a clear scaling trend where larger models produce more informative entropy signals. A key finding is that entropy primarily captures knowledge uncertainty rather than reasoning difficulty: the signal is strongest for factual recall domains such as Biology and Economics, and weakest for reasoning-intensive domains such as Math and Law. The Model-as-Judge approach, by contrast, performs near-randomly (ROC-AUC ≈ 0.49), indicating that external evaluation captures fundamentally different properties than the model's own internal uncertainty. Calibration analysis revealed systematic overconfidence across all architectures, particularly in high-certainty regimes, suggesting that entropy-based monitoring could serve as a valuable safeguard in deployment. This work was recognized with the Best Paper Award at AINL 2025.

In the second study, we discovered that language-specific directions can be reliably identified in LLM latent space through PCA, achieving 96–100% classification accuracy across four language pairs on Qwen2.5-1.5B and Llama-3.2-1B. The layer-wise analysis revealed that language identity crystallizes in final layers while early layers focus on semantic processing, suggesting these properties occupy largely orthogonal subspaces. By projecting out the source language component from hidden states during inference, we reduced distributional divergence by 25–55% and achieved 63–95% reduction in Code-Switching Index in generated text, without requiring any additional training. The method demonstrates that behavioral control can be achieved through geometric manipulation of internal representations, though the fragility of steering coefficients—where small changes in magnitude can shift output from fluent text to degenerate repetition—represents an important practical constraint that requires careful per-language calibration.

In the third study, we developed a training pipeline that uses single-token answer entropy to stratify training data by complexity and applies targeted interventions: direct supervised fine-tuning for regular examples and chain-of-thought distillation from a teacher model ensemble for hard examples. This approach achieved accuracy improvements from 0.39 to 0.52 (Qwen-3B) and 0.51 to 0.64 (Phi-4-Mini) on MMLU-Pro while using 79–82% less training data than full distillation. The method generalizes across three datasets (MMLU-Pro, GSM8K, MedMCQA) and three model architectures, with particularly strong gains on mathematical reasoning (GSM8K: 0.82 vs. 0.79 for full distillation). The sensitivity analysis confirmed that single-token entropy substantially outperforms both MASJ-based metrics and chain-of-thought entropy variants for complexity estimation, and that standard curriculum learning is fragile and insufficient without differentiated training strategies.

8.2 Unifying Insights

The three studies share a common finding: the signals produced during a single forward pass through an LLM encode rich information that extends well beyond the immediate task of next-token prediction. Output distributions reflect model confidence in a domain-dependent manner, with high entropy indicating uncertainty and the relationship between entropy and correctness being strongest for knowledge-intensive questions where the model either possesses relevant training data or does not. Internal representations encode interpretable geometric structure, with language identity represented as linear directions in activation space that emerge progressively through layers and concentrate in the final transformer blocks. The model's uncertainty about a training example, as measured by output entropy, reflects the example's intrinsic difficulty in a way that is actionable for training—different complexity levels genuinely benefit from different training interventions rather than merely different ordering.

These insights suggest that LLMs develop internal representations of their own capabilities and limitations that can be accessed and exploited without explicit training for self-assessment. The forward pass produces, as a byproduct of next-token prediction, signals about confidence, language identity, and input complexity that are linearly accessible and practically useful. This observation has implications beyond the three specific applications explored here: any property that is linearly encoded in the model's representations is potentially amenable to similar extraction and manipulation techniques.

A particularly interesting connection emerges between Studies 1 and 3. Both exploit the same signal—output entropy—but for different purposes. In Study 1, entropy serves as a runtime indicator of answer reliability, enabling users to decide whether to trust a given response. In Study 3, the same entropy signal is used offline during training to allocate computational resources efficiently. This dual use of a single signal demonstrates the versatility of forward-pass information and suggests that a unified monitoring and training framework could leverage entropy for both deployment-time quality assurance and training-time data curation simultaneously.

A deeper unifying theme connects all three studies: at their core, they all explore uncertainty and the various ways it can be measured and exploited. The first study measures uncertainty directly through output entropy. The second identifies geometric structure in the latent space that separates language identity from semantic content. The third uses uncertainty as a proxy for example complexity to guide training. What ties these together is the prospect of *uncertainty decomposition*. The language directions identified in Study 2 define an orthogonal decomposition of the representation space into a language-specific subspace and a language-agnostic complement. If uncertainty signals from Study 1 were analyzed along this decomposition, one could potentially disentangle language-specific uncertainty (e.g., the model is unsure which language to generate in) from language-agnostic uncertainty (e.g., the model genuinely lacks knowledge about the topic). Such a decomposition would provide more actionable uncertainty estimates: language-specific uncertainty could be resolved by steering, while language-agnostic uncertainty would signal a genuine knowledge gap requiring external intervention. This perspective suggests that the methods developed in this thesis are not three independent techniques but rather complementary components of a broader framework for structured uncertainty analysis in LLMs.

8.3 Position in the Research Landscape

Our work contributes to several active research areas. In the domain of uncertainty quantification, we extend prior work on semantic entropy [28] and confidence calibration [16] by demonstrating that even the simplest entropy measure—applied to a single answer token—provides strong predictive signal when analyzed at the appropriate granularity (per domain, per reasoning type). Our

approach requires only a single forward pass, making it practical for real-time applications where ensemble methods or Monte Carlo dropout would be prohibitively expensive.

In interpretability and steering, we build on discoveries about linear structure in LLM representations [39, 46] and extend the concept erasure framework [43, 3] to active generation control. While prior work in representation engineering [55] focused on concepts such as honesty and harmlessness, we demonstrate that language identity provides a particularly clean test case for activation steering, with near-perfect linear separability and clear practical motivation. Our finding that optimal steering requires active counter-steering beyond mere projection removal ($|s| > 1$) adds nuance to the linear representation hypothesis.

In efficient training, our complexity-aware approach relates to curriculum learning [4] and data selection [44], but differs in a critical way: rather than simply ordering examples by difficulty, we apply fundamentally different training procedures to different complexity tiers. The insight that letting the model focus on what it can already solve directly, while giving extra guidance only where it struggles, yields a better allocation of limited model capacity than either uniform SFT or uniform distillation.

8.4 Limitations

Several limitations should be acknowledged, organized by the study they primarily affect.

Regarding model scale and diversity, the uncertainty estimation study uses models from 1.5B to 72B parameters, but the language steering and fine-tuning experiments are limited to small models in the 1–3B range. Whether the geometric properties of language directions and the effectiveness of entropy-based stratification transfer to models with 70B+ parameters or Mixture-of-Experts architectures remains an open question. The strong scaling trend observed in Study 1 suggests that larger models may produce even more useful signals, but this requires empirical validation.

Regarding language and task coverage, while language steering was validated on four language pairs spanning different typological distances (English–Chinese, English–Spanish, English–Russian, English–Hindi), low-resource languages, right-to-left scripts, and agglutinative or tonal languages remain unexplored. English–Hindi consistently showed the weakest improvements across all metrics, suggesting that typologically distant pairs with different scripts present additional challenges. On the task side, all three studies focus on question-answering settings; the applicability of entropy-based uncertainty estimation and complexity-aware fine-tuning to open-ended generation, coding tasks, or dialogue systems needs investigation.

The steering fragility observed in Study 2 represents a practical limitation. Optimal coefficients require manual grid search per language pair and depend on whether the optimization target is distributional similarity or generation quality. Small changes in coefficient magnitude can cause transitions from fluent output to degenerate repetition, and no principled method for automatic coefficient selection has been developed. The artificial code-switching evaluation setup, where the second half of each sentence is machine-translated, may not capture the full complexity of naturally occurring code-switching patterns at morphosyntactic boundaries.

For the complexity-aware fine-tuning pipeline, low entropy can still correspond to hallucinations, leading to imperfect identification of question complexity. The three-way equal split into easy, medium, and hard tiers was chosen for simplicity and may not be optimal for all datasets. In settings where a teacher model is unavailable or provides low-quality reasoning chains, the benefit of chain-of-thought distillation for hard examples would be diminished.

8.5 Future Directions

This research opens several promising directions for future investigation. The three types of signals explored in this thesis—entropy patterns, geometric structure of representations, and complexity scores—could be combined into unified frameworks for model monitoring and control. A deployment system that simultaneously monitors answer confidence via entropy, enforces language consistency via steering, and continuously improves via complexity-aware fine-tuning would represent a practical step toward more reliable LLM applications.

The methods could be extended to enable online adaptation during deployment. Rather than computing complexity scores offline on a fixed dataset, entropy could be monitored in real time to identify examples where the model is struggling, with automatic escalation to a larger model or human review. Similarly, language steering coefficients could be adapted dynamically based on observed code-switching patterns in production traffic.

Beyond language identity, the geometric structure of latent space could be exploited for steering other attributes such as formality, domain expertise, safety compliance, or reasoning style. The near-perfect linear separability we observed for language suggests that other high-level attributes may also be linearly encoded, making multi-dimensional steering a realistic prospect. Developing theoretical foundations for understanding why these signals are predictive—connecting entropy to information-theoretic notions of model knowledge and linear directions to learned feature representations—would strengthen the basis for future applications and help predict when these methods will succeed or fail.

A particularly promising direction is structured uncertainty decomposition. The language directions identified in Study 2 define orthogonal subspaces in the representation space; projecting entropy signals from Study 1 onto these subspaces could separate language-specific uncertainty from language-agnostic uncertainty. Such decomposition would make uncertainty estimates more actionable: language-specific uncertainty could be resolved cheaply via steering, while language-agnostic uncertainty would signal a genuine knowledge gap requiring external intervention or abstention.

Finally, integrating the uncertainty and control mechanisms developed here with reinforcement learning from human feedback and other alignment techniques could improve model safety. Entropy-based confidence scores could modulate the strength of safety interventions, applying stronger guardrails when the model is uncertain and allowing more freedom when it is confident. This adaptive approach could reduce the tension between helpfulness and safety that characterizes current alignment methods.

8.6 Conclusion

This thesis demonstrated that accessible signals from a single forward pass through large language models provide valuable information for addressing fundamental deployment challenges. Single-token answer entropy predicts answer correctness with ROC-AUC up to 0.83 in knowledge-dependent domains, latent space geometry enables language control with 63–95% reduction in code-switching, and complexity-aware data stratification improves training efficiency while using 79–82% less data. Together, these contributions establish that exploiting forward-pass signals offers a practical path toward more reliable, controllable, and efficient LLMs—without requiring architectural modifications or expensive retraining.

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Innovations

This chapter describes the practical applications and innovation potential of the research results presented in this thesis.

Industrial Applications of Research Results

1. **Real-time reliability monitoring for LLM deployments.** The uncertainty estimation pipeline developed in Study 1 enables automatic detection of unreliable LLM outputs at inference time, requiring only a single forward pass with no additional models or computational overhead. This is directly applicable to high-stakes domains:

   - *Healthcare*: Flagging uncertain diagnostic suggestions before they reach clinicians
   - *Legal*: Identifying potentially hallucinated legal citations or case references
   - *Education*: Warning students when AI-generated explanations may be unreliable
   - *Customer support*: Escalating uncertain responses to human agents

   The finding that entropy-based uncertainty is domain-dependent (effective for knowledge questions, less so for reasoning) provides actionable guidance for deploying such systems: operators can calibrate confidence thresholds per domain.

2. **Multilingual chatbot language control.** The language steering method from Study 2 addresses a concrete industry pain point: multilingual chatbots that unintentionally switch languages mid-response, degrading user experience. The method offers several advantages for industrial deployment:

   - *Zero training cost*: Steering requires only parallel translations for PCA calibration (as few as 50 sentence pairs), with no fine-tuning
   - *Negligible inference overhead*: One dot product and one vector subtraction per token
   - *No model modification*: Works with any decoder-only LLM via forward hooks

   Potential applications include customer service platforms, content localization pipelines, and real-time translation assistants.

3. **Cost-efficient domain adaptation.** The complexity-aware fine-tuning pipeline from Study 3 reduces the computational cost of adapting LLMs to specific domains by 79–82% in terms of training data (tokens processed), while maintaining or improving accuracy. This has direct implications for:

   - *Enterprise AI*: Companies can fine-tune domain-specific models (medical, legal, financial) at a fraction of the current cost
   - *Resource-constrained settings*: Organizations with limited GPU budgets can achieve competitive performance
   - *Rapid iteration*: Faster training cycles enable more frequent model updates as domain knowledge evolves

Open-Source Contributions

All three research projects have released their code and data publicly to facilitate reproducibility and further research:

1. Uncertainty estimation pipeline: `github.com/LabARSS/question-complextiy-estimation`
2. Complexity-aware fine-tuning: `github.com/LabARSS/complexity-aware-fine-tuning`
3. Language steering: `github.com/fxlrnrpt/language-steering-in-latent-space`

Start-up Potential

The combination of the three methods developed in this thesis—uncertainty monitoring, language control, and efficient fine-tuning—forms a coherent product offering for an "LLM reliability and control" platform. Such a platform could provide:

- Automated confidence scoring for LLM outputs (per-domain calibrated)
- Language consistency enforcement for multilingual deployments
- Efficient domain adaptation services with reduced compute requirements

All methods operate on standard transformer architectures without architectural modifications, making them compatible with the most widely deployed open-source LLMs.

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Appendix

A. Prompts for Uncertainty Estimation (Study 1)

The following prompts were used for complexity estimation in the MMLU-Pro evaluation pipeline.

A.1 Single-Token Response Prompts

Standard prompt:

The following are multiple choice questions about subject.
Write down ONLY the NUMBER of the correct answer and nothing else.

With fallback for unknown answers:

The following are multiple choice questions about subject.
If you are certain about the answer return the correct option number, otherwise return 0. Write down ONLY the NUMBER and nothing else.

A.2 Chain-of-Thought Response Prompts

Standard CoT prompt:

The following are multiple choice questions about subject. Explain your thinking process step-by-step. At the end, write down the number of the correct answer by strictly following this format: [number of correct answer].

CoT with fallback:

The following are multiple choice questions about subject. Explain your thinking process step-by-step. At the end, if you are certain about the answer write down the number of the correct answer by strictly following this format: [number of correct answer], otherwise return [0].

A.3 MASJ Prompts

Reasoning requirement estimation:

You are an expert in the topic of the question. Please act as an impartial judge and evaluate the complexity of the multiple-choice question with options below. Begin your evaluation by providing a short explanation. Be as objective as possible. After providing your explanation, you must rate the question complexity by strictly following the criteria: [Number of reasoning steps] – how many reasoning steps do you need to answer this question? Valid answers: low, medium, high.

Education level estimation:

You are an expert in the topic of the question. Please act as an impartial judge and evaluate the complexity of the multiple-choice question with options below. After providing your explanation, you must rate the question complexity by strictly following the scale: "high school and easier", "undergraduate", "graduate", "postgraduate".

B. Training Hyperparameters (Study 3)

Table 8.1: Training hyperparameters for MMLU-Pro experiments.

Parameter	SFT	Pipeline (easy/mid)	Pipeline (hard)	Distillation
Effective batch size	64	64	64	64
Max new tokens	30	30	8192	8192
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning Rate	1e-4	1e-4	1e-4	1e-4

Table 8.2: Training hyperparameters for GSM8K/MedMCQA experiments.

Parameter	SFT	Pipeline (easy/mid)	Pipeline (hard)	Distillation
Effective batch size	256	256	256	256
Max new tokens	100	100	8192	8192
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning Rate	1e-4	1e-4	1e-4	1e-4
LoRA rank	64	64	64	64
LoRA alpha	128	128	128	128
LoRA dropout	0.05	0.05	0.05	0.05

C. Steering Coefficients (Study 2)

Table 8.3: Llama-3.2-1B steering coefficients: KL-optimized (from distributional analysis) vs. CSI-optimized (from generation grid search).

Language pair	KL-opt.	CSI-opt.
English – Chinese	-5.0	-5.0
English – Spanish	-3.7	-5.0
English – Russian	-3.5	-4.0
English – Hindi	-3.6	-5.0

CSI-optimized coefficients tend toward higher magnitudes, which improves language control but increases degeneration risk. The coefficients cluster near the maximum tested value (-5.0), suggesting that direct optimization of generation metrics pushes toward an aggressive regime where language control and fluency trade off sharply.

D. Supplementary Distributional Metrics (Study 2)

Table 8.4: JS divergence (no steering → steering, lower is better).

Language pair	Qwen	Llama
English – Chinese	0.67 → 0.49	0.65 → 0.51
English – Spanish	0.59 → 0.52	0.58 → 0.44
English – Russian	0.62 → 0.50	0.62 → 0.43
English – Hindi	0.64 → 0.60	0.63 → 0.54

Table 8.5: Top-50 token overlap (no steering → steering, higher is better).

Language pair	Qwen	Llama
English – Chinese	3.69% → 26.46%	4.47% → 24.31%
English – Spanish	11.41% → 20.23%	11.45% → 30.74%
English – Russian	7.41% → 25.45%	7.49% → 31.96%
English – Hindi	11.23% → 11.68%	6.54% → 25.88%